Calculus Notes

Contents

Question: What is Calculus?
Answer: Calculus is the mathematics of change.

Calculus provides the tools to study quantities that vary continuously. Questions such as how fast a car is moving at a given instant, how the temperature of a room changes, or how engineers design efficient structures all involve continuous change, and calculus provides a mathematical framework for analyzing them.

Calculus is widely used in science, engineering, economics, computer science, and data science. It plays a fundamental role in modeling physical phenomena, optimization, and the analysis of dynamic systems.

Mathematically, calculus is the study of limits, continuity, derivatives, integrals, sequences, and infinite series. It was developed independently by **Sir Isaac Newton** (1642–1727) and **Gottfried Wilhelm Leibniz** (1646–1716) in the late seventeenth century.

Sequence of Real Numbers

A sequence of real numbers is an infinite ordered list of real numbers indexed by the natural numbers. In this section, we introduce the concept of convergence and study the fundamental properties of convergent sequences. We also discuss monotone sequences and the Monotone Convergence Theorem. The section concludes with the Cauchy criterion, which characterizes convergence without requiring prior knowledge of the limit.

Series of Real Numbers

A series is the sum of the terms of a sequence. In this section, we study infinite series and the conditions under which they converge. We develop several convergence tests and illustrate their applications. Infinite series play a fundamental role in mathematics and have numerous applications in science, engineering, and computer science.

1 Basics

Definition 1.1. *Let $A \subset \mathbb{R}$ be nonempty. A real number α is called an* ***upper bound*** *of A if for each $x \in A$, we have*

$$x \leq \alpha.$$

In terms of quantifiers:

$$\forall x \in A \ (x \leq \alpha).$$

Geometrically, this means that every element of A lies to the left of or at α on the number line.

Example 1.1. *Let*

$$A = (1, 4).$$

Then 5 is an upper bound of A. In fact, every real number greater than or equal to 4 is an upper bound of A.

Definition 1.2. *Let $A \subset \mathbb{R}$ be nonempty. We say that A is* ***bounded above*** *in $\mathbb{R}$ if there exists an upper bound of A. Equivalently, there exists $\alpha \in \mathbb{R}$ such that*

$$x \leq \alpha, \qquad \forall x \in A.$$

In terms of quantifiers:

$$\exists \alpha \in \mathbb{R}\ (\forall x \in A\ (x \leq \alpha)).$$

Example 1.2. *The interval*

$$A = (-2, 3]$$

is bounded above because 3 *is an upper bound of* A*.*

Definition 1.3. *Let* $A \subset \mathbb{R}$ *be nonempty. We say that* A *is* ***not bounded above*** *in* $\mathbb{R}$ *if for every* $\alpha \in \mathbb{R}$*, there exists* $x \in A$ *such that*

$$x > \alpha.$$

In terms of quantifiers:

$$\forall \alpha \in \mathbb{R}\ (\exists x \in A\ (x > \alpha)).$$

Example 1.3. *The interval*

$$A = (0, \infty)$$

is not bounded above since, for every real number α*, there exists* $x \in (0, \infty)$ *with* $x > \alpha$*.*

Definition 1.4. *Let* $A \subset \mathbb{R}$ *be nonempty. A real number* β *is called a* ***lower bound*** *of* A *if*

$$\beta \leq x, \qquad \forall x \in A.$$

Equivalently,

$$\forall x \in A\ (\beta \leq x).$$

Example 1.4. *Let*

$$A = (2, 6).$$

Then 1 *is a lower bound of* A*. In fact, every real number less than or equal to* 2 *is a lower bound of* A*.*

Definition 1.5. *A nonempty set* $A \subset \mathbb{R}$ *is said to be* ***bounded below*** *if there exists a real number* β *such that*

$$\beta \leq x, \qquad \forall x \in A.$$

Equivalently,

$$\exists\, \beta \in \mathbb{R}\ (\forall x \in A\ (\beta \leq x)).$$

Example 1.5. *The interval*

$$A = [-3, 2)$$

is bounded below because -3 *is a lower bound of* A*.*

Definition 1.6. *Let $A \subset \mathbb{R}$ be nonempty. We say that A is **not bounded below** in $\mathbb{R}$ if for every $\beta \in \mathbb{R}$, there exists $x \in A$ such that*

$$x < \beta.$$

In terms of quantifiers:

$$\forall \beta \in \mathbb{R}\ (\exists x \in A\ (x < \beta)).$$

Example 1.6. *The interval*

$$A = (-\infty, 1)$$

is not bounded below since, for every real number β, there exists $x \in (-\infty, 1)$ with $x < \beta$.

Definition 1.7. *A set $A \subset \mathbb{R}$ is called **bounded** if it is bounded above and bounded below. Equivalently, there exist real numbers m and M such that*

$$m \leq x \leq M, \qquad \forall x \in A.$$

Example 1.7. *The interval*

$$A = (-1, 4)$$

is bounded because every element $x \in A$ satisfies

$$-1 < x < 4.$$

Hence, -1 is a lower bound and 4 is an upper bound of A.

Definition 1.8. *Let $A \subset \mathbb{R}$ be a nonempty set that is bounded above. A real number $\alpha \in \mathbb{R}$ is called the **least upper bound (LUB)** or **supremum** of A if*

1. *α is an upper bound of A; and*

2. *if β is any upper bound of A, then $\alpha \leq \beta$.*

We write

$$\alpha = \sup A.$$

Example 1.8. *Let*

$$A = (0, 1).$$

Then

$$\sup A = 1.$$

Notice that $1 \notin A$. Thus, the supremum of a set need not belong to the set.

Definition 1.9. *Let $A \subset \mathbb{R}$ be a nonempty set that is bounded below. A real number $\alpha \in \mathbb{R}$ is called the **greatest lower bound (GLB)** or **infimum** of A if*

1. *α is a lower bound of A; and*

2. *if β is any lower bound of A, then $\beta \leq \alpha$.*

We write
$$\alpha = \inf A.$$

Example 1.9. *Let*
$$A = (2, 5).$$
Then
$$\inf A = 2.$$
Notice that $2 \notin A$. Thus, the infimum of a set need not belong to the set.

Example 1.10. *Let $A = [0, 1]$. Then $\sup A = 1,\ \inf A = 0$. Since both endpoints belong to the interval, the supremum and infimum are elements of the set.*

Theorem 1.11 (Least Upper Bound Property). *Every nonempty subset of $\mathbb{R}$ that is bounded above possesses a least upper bound (supremum) in $\mathbb{R}$.*

Theorem 1.12 (Greatest Lower Bound Property). *Every nonempty subset of $\mathbb{R}$ that is bounded below possesses a greatest lower bound (infimum) in $\mathbb{R}$.*

1.1 Properties of the Supremum and the Infimum

Theorem 1.13 (Properties of the Supremum). *Let S be a nonempty subset of $\mathbb{R}$ that is bounded above. Then the supremum of S exists. Let*
$$M = \sup S.$$
Then $M \in \mathbb{R}$ and satisfies the following properties:

1. *Every element of S is less than or equal to M; that is,*
$$x \in S \implies x \leq M.$$
2. *For every $\varepsilon > 0$, there exists an element $y(\varepsilon) \in S$ such that*
$$M - \varepsilon < y(\varepsilon) \leq M.$$

Theorem 1.14 (Properties of the Infimum). *Let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Then the infimum of S exists. Let*
$$m = \inf S.$$
Then $m \in \mathbb{R}$ and satisfies the following properties:

1. *Every element of S is greater than or equal to m; that is,*
$$x \in S \implies x \geq m.$$
2. *For every $\varepsilon > 0$, there exists an element $y(\varepsilon) \in S$ such that*
$$m \leq y(\varepsilon) < m + \varepsilon.$$

Remark 1.15. *The notation $y(\varepsilon)$ indicates that the choice of the element y depends on the value of ε.*

Example 1.16. *Prove that the set $\mathbb{N}$ is not bounded above.*

Proof. The set $\mathbb{N}$ is nonempty since $1 \in \mathbb{N}$. Suppose, to the contrary, that $\mathbb{N}$ is bounded above. Then, by the Least Upper Bound Property of $\mathbb{R}$, the set $\mathbb{N}$ has a supremum. Let $u = \sup \mathbb{N}$. Since u is the supremum of $\mathbb{N}$, it satisfies the following properties:

1. $n \leq u$ for every $n \in \mathbb{N}$.

2. For every $\varepsilon > 0$, there exists $k \in \mathbb{N}$ such that $u - \varepsilon < k \leq u$.

Choose $\varepsilon = 1$. Then there exists $k \in \mathbb{N}$ such that

$$u - 1 < k \leq u.$$

Adding 1 throughout the inequality gives

$$u < k + 1.$$

Since $k + 1 \in \mathbb{N}$, we have found a natural number greater than u, contradicting the fact that u is an upper bound of $\mathbb{N}$. Hence our assumption is false. Therefore,

$$\mathbb{N} \text{ is not bounded above.}$$

$\square$

Theorem 1.17 (**Archimedean Property of $\mathbb{R}$**). *If $x, y \in \mathbb{R}$ with $x > 0$ and $y > 0$, then there exists a natural number n such that*

$$ny > x.$$

Corollary 1.18. *The following statements are consequences of the Archimedean Property.*

1. If $x \in \mathbb{R}$, then there exists $n \in \mathbb{N}$ such that
$$n > x.$$

2. If $x > 0$, then there exists $n \in \mathbb{N}$ such that
$$0 < \frac{1}{n} < x.$$

3. If $x > 0$, then there exists $m \in \mathbb{Z}$ such that
$$m - 1 \leq x < m.$$

4. If $x \in \mathbb{R}$, then there exists an integer $m \in \mathbb{Z}$ such that
$$m - 1 \leq x < m.$$

Theorem 1.19 (**Density Property of the Rational Numbers**). *If $x, y \in \mathbb{R}$ with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that*

$$x < r < y.$$

Theorem 1.20 (**Density Property of the Irrational Numbers**). *If $x, y \in \mathbb{R}$ with $x < y$, then there exists an irrational number $s \in \mathbb{R} \setminus \mathbb{Q}$ such that*

$$x < s < y.$$

Definition 1.10 (Absolute Value). *For $x \in \mathbb{R}$, the* absolute value *of x is defined by*

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Remark 1.21. *We now recall some useful properties of the absolute value. For all $a, b \in \mathbb{R}$, the following properties hold:*

1. $|a| \geq 0$, $|a| = 0 \iff a = 0$. *Moreover,* $-|a| \leq a \leq |a|$.
2. *Let $\varepsilon > 0$. Then*
$$|a - b| < \varepsilon \iff a \in (b - \varepsilon,\ b + \varepsilon).$$
3. $$|ab| = |a||b|,$$
and, provided $b \neq 0$,
$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$
4. *(Triangle Inequality)*
$$|a + b| \leq |a| + |b|.$$
5. $$\big||a| - |b|\big| \leq |a - b|.$$
6. $$\max\{a, b\} = \frac{a + b + |a - b|}{2}, \qquad \min\{a, b\} = \frac{a + b - |a - b|}{2}.$$

2 Sequences

Sequences arise naturally in the study of approximation. For example,

$$\frac{1}{7} = 0.142857142857\ldots$$

can be approximated by the sequence

$$0.1,\ 0.14,\ 0.142,\ 0.1428,\ 0.14285,\ \ldots$$

obtained by retaining one additional decimal place at each step. As more decimal places are kept, the approximation becomes more accurate. For instance, any term with at least four decimal places approximates $\frac{1}{7}$ with an error less than 10^{-4}.

The concept of convergence provides a rigorous way to describe this idea. A sequence (a_n) is said to converge to a real number x if the terms a_n become arbitrarily close to x as n becomes sufficiently large.

Definition 2.1. *A sequence of real numbers is a function from the set $\mathbb{N}$ of natural numbers to the set $\mathbb{R}$ of real numbers. If $f : \mathbb{N} \to \mathbb{R}$ is a sequence, and if $a_n = f(n)$ for $n \in \mathbb{N}$, then we write the sequence f as (a_n) or $(a_1, a_2, \ldots)$.*

Example 2.1. *Some examples of sequences are given below.*

1. *Fix* $c \in \mathbb{R}$ *and define* $a_n = c$ *for all* $n \in \mathbb{N}$. *Then* $(a_n) = (c, c, c, \ldots)$. *This is called a constant sequence.*
2. *Let* $a_n = \frac{1}{n}$. *Then* $(a_n) = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots\right)$.
3. *Let* $a_n = \frac{(-1)^{n+1}}{n}$. *Then* $(a_n) = \Big(1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \ldots, \frac{1}{2k-1}, -\frac{1}{2k}, \ldots\Big)$.
4. *Let* $a_n = (-1)^n$. *Then* $(a_n) = (-1, 1, -1, 1, \ldots)$.
5. *Let* $a_n = n$. *Then* $(a_n) = (1, 2, 3, 4, \ldots)$.
6. *Let* $a_n = 2^n$. *Then* $(a_n) = (2, 4, 8, 16, \ldots)$.
7. *Let* $a_n = \frac{1}{2^n}$. *Then* $(a_n) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots\right)$.
8. *Let* $a_1 = a_2 = 1$ *and define* $a_n = a_{n-1} + a_{n-2}$ *for all* $n \geq 3$. *Then* $(a_n) = (1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots)$.

Remark 2.2. *A sequence* (a_n) *is different from the set* $\{a_n : n \in \mathbb{N}\}$. *In a sequence, the order of terms and repetitions matter, whereas in a set they do not. For example, the sequence*

$$(1, \tfrac{1}{2}, 1, \tfrac{1}{3}, \ldots, 1, \tfrac{1}{n+1}, \ldots),$$

defined by $a_{2n-1} = 1$ *and* $a_{2n} = \frac{1}{n+1}$, *contains infinitely many repetitions of* 1, *but*

$$\{a_n : n \in \mathbb{N}\} = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}.$$

Example 2.3. *Let*

$$x_1 = 0.1, \quad x_2 = 0.14, \quad x_3 = 0.142, \quad x_4 = 0.1428, \quad \ldots,$$

where a_n *is obtained by retaining the first* n *digits after the decimal point in the decimal expansion of*

$$\frac{1}{7} = 0.142857142857\ldots$$

Observe that

$$\left|\frac{1}{7} - a_n\right| < \frac{1}{10^n}, \qquad \forall n \in \mathbb{N}.$$

Thus, given any prescribed level of accuracy, we can choose n *sufficiently large so that* a_n *approximates* $\frac{1}{7}$ *to that accuracy. This example motivates the notion of convergence, which we study next.*

2.1 Convergence and divergence

A central concept in analysis is that of *convergence*. We begin by studying the convergence of sequences. Consider the sequences and observe their behavior as n increases:

1. $a_n = 1$: every term is equal to 1.
2. $a_n = n$: the terms increase without bound.
3. $a_n = \frac{1}{n}$: the terms approach 0.
4. $a_n = \frac{n}{n+1}$: the terms approach 1.
5. $a_n = (-1)^n$: the terms alternate between -1 and 1, and do not approach any real number.

We now give a precise mathematical meaning to the statement that a sequence (a_n) *approaches* a real number a as n becomes arbitrarily large.

Definition 2.2. *Let* (a_n) *be a sequence of real numbers and let* $a \in \mathbb{R}$. *We say that* (a_n) ***converges*** *to* a *if for every* $\epsilon > 0$, *there exists* $N \in \mathbb{N}$ *such that*

$$|a_n - a| < \epsilon, \qquad \forall\, n \geq N.$$

The number a *is called the* ***limit*** *of* (a_n). *We write* $a_n \to a \quad$ *as* $n \to \infty$, *or equivalently,*

$$\lim_{n\to\infty} a_n = a.$$

Remark 2.4.

1. *In the definition of convergence, the number* $\epsilon > 0$ *is arbitrary, however small. For each such* ϵ, *there exists* $N \in \mathbb{N}$ *such that*

$$|a_n - a| < \epsilon, \qquad \forall\, n \geq N.$$

2. *Equivalently, for every* $\epsilon > 0$, *there exists* $N \in \mathbb{N}$ *such that*

$$a_n \in (a - \epsilon, a + \epsilon), \qquad \forall\, n \geq N.$$

Thus, every ϵ*-neighborhood of* a *contains all but finitely many terms of the sequence.*

3. *Suppose* (a_n) *is a sequence and* $a \in \mathbb{R}$. *Then to show that* (a_n) *does not converge to* a, *we should be able to find an* $\epsilon > 0$ *such that infinitely many* a_n*'s are outside the interval* $(a - \epsilon, a + \epsilon)$.

Example 2.5.

1. *Consider the sequence* $\left(\frac{1}{n}\right)_n$, *where* $a_n = \frac{1}{n}$. *We claim that*

$$\lim_{n\to\infty} a_n = 0.$$

Let $\epsilon > 0$ *be given. We have*

$$|a_n - 0| = \frac{1}{n}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ *such that* $N > \frac{1}{\epsilon}$, *Hence, for every* $n \geq N$,

$$|a_n - 0| = \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Therefore, $\lim\limits_{n\to\infty} a_n = 0$.

2. *Consider the sequence* $a_n = \frac{n^2+1}{n^2}$. *We claim that* $\lim\limits_{n\to\infty} a_n = 1$. *Let* $\epsilon > 0$ *be given. We have*

$$|a_n - 1| = \left|\frac{n^2+1}{n^2} - 1\right| = \frac{1}{n^2}.$$

By the Archimedean Property, there exists $N \in \mathbb{N}$ *such that* $N > \frac{1}{\sqrt{\epsilon}}$. *Hence, for every* $n \geq N$,

$$|a_n - 1| = \frac{1}{n^2} \leq \frac{1}{N^2} < \epsilon.$$

Therefore, $\lim\limits_{n\to\infty} a_n = 1$.

3. *The sequence* $\left(1 - \frac{1}{n}\right)_n$ *converges to* 1. *Let* $a_n = 1 - \frac{1}{n}$. *Since*

$$|a_n - 1| = \frac{1}{n},$$

the same choice of N yields

$$|a_n - 1| < \epsilon, \qquad n \geq N.$$

Therefore, $\lim_{n\to\infty} a_n = 1$.

The preceding examples illustrate the basic strategy for proving convergence. We first identify a candidate for the limit and then use the definition of convergence to show that the sequence approaches it. This naturally leads to the following questions:

1. How can we determine the limit of a sequence?

2. Is the limit of a sequence unique?

The first question is often answered by experience and the algebraic properties of limits. We now address the second question.

Theorem 2.6 (**Uniqueness of the Limit**). *If*

$$\lim_{n\to\infty} a_n = x \quad \text{and} \quad \lim_{n\to\infty} a_n = y,$$

then $x = y$.

Proof. Let $\epsilon > 0$ be given. Since $a_n \to x$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - x| < \frac{\epsilon}{2}, \qquad \forall n \geq N_1.$$

Similarly, since $a_n \to y$, there exists $N_2 \in \mathbb{N}$ such that

$$|a_n - y| < \frac{\epsilon}{2}, \qquad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$, the triangle inequality gives

$$|x - y| = |x - a_n + a_n - y| \leq |x - a_n| + |a_n - y| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $\epsilon > 0$ is arbitrary, it follows that $x = y$. ☐

Example 2.7. *1. Consider the sequence $a_n = n$. We claim that this sequence does not converge. Assume, on the contrary, that (a_n) converges to some $a \in \mathbb{R}$. Following the definition of convergence, the interval $(a-1, a+1)$ must contain all but finitely many terms of the sequence. Hence, there exists $N \in \mathbb{N}$ such that*

$$n = a_n \in (a-1, a+1), \qquad \forall n \geq N.$$

In particular, $n < a+1$ for all $n \geq N$. Thus, the set $\mathbb{N}$ is bounded above by $\max\{N, a+1\}$, contradicting the fact that $\mathbb{N}$ is not bounded above. Therefore, (a_n) does not converge.

2. Consider the sequence $a_n = (-1)^n$. Observe that the intervals

$$(0, \infty) \quad \text{and} \quad (-\infty, 0)$$

each contain infinitely many terms of the sequence. Hence, no interval containing a possible limit can contain all but finitely many terms of the sequence. Therefore, $((-1)^n)$ is not convergent.

Remark 2.8. *A sequence that is not convergent is called* **divergent**. *A divergent sequence need not tend to infinity; it may oscillate, as in $\big((-1)^n\big)$, or increase without bound, as in (n). Thus, a sequence can fail to converge in different ways. To prove that a sequence is divergent, it suffices to show that it does not converge to any real number.*

2.2 Properties of convergent sequences

Definition 2.3. *A sequence* (a_n) *of real numbers is said to be* ***bounded*** *if there exists a constant* $C > 0$ *such that*

$$|a_n| \leq C, \qquad \forall n \in \mathbb{N}.$$

Example 2.9. *The following are some examples of bounded and unbounded sequences.*

1. *Every constant sequence is bounded.*
2. *The sequence* $\left(\frac{1}{n}\right)$ *is bounded.*
3. *The sequence* $((-1)^n)$ *is bounded.*
4. *The sequence* $\left(\frac{n^2+1}{n^2}\right)$ *is bounded.*
5. *The sequence* $(\sin n)$ *is bounded.*

Definition 2.4. *A sequence* (a_n) *of real numbers is said to be* ***unbounded*** *if, for every* $C > 0$*, there exists* $n \in \mathbb{N}$ *such that* $|a_n| > C$*.*

Example 2.10. *The sequences* $(n), (n^2), (n^n)$ *are unbounded.*

Theorem 2.11. *Every convergent sequence of real numbers is bounded.*

Proof. Let (a_n) be a sequence converging to $a \in \mathbb{R}$. Choose $\epsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < 1, \qquad \forall n \geq N.$$

By the triangle inequality,

$$|a_n| = |a_n - a + a| \leq |a_n - a| + |a| < 1 + |a|, \qquad \forall n \geq N.$$

Now let

$$C = \max\{|a_1|, \ldots, |a_{N-1}|, 1 + |a|\}.$$

Then

$$|a_n| \leq C, \qquad \forall n \in \mathbb{N}.$$

Hence, (a_n) is bounded. $\square$

Example 2.12. *Converse of the previous theorem is not true. For* $n \in \mathbb{N}$*, let*

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}.$$

Then (a_n) *diverges: To see this, observe that*

$$\begin{aligned} a_{2n} &= 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{2n} \\ &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \\ &\quad \ldots + \left(\frac{1}{2^{n-1}+1} + \ldots + \frac{1}{2^n}\right) \\ &\geq 1 + \frac{n}{2}. \end{aligned}$$

Hence, (a_n) *is not a bounded sequence, so that it diverges.*

Theorem 2.13. *Let (a_n) be a sequence of real numbers such that*

$$\lim_{n\to\infty} a_n = a,$$

where $a \neq 0$. Then there exists $N \in \mathbb{N}$ such that

$$|a_n| > \frac{|a|}{2}, \qquad \forall n \geq N.$$

Proof. Choose $\epsilon = \frac{|a|}{2} > 0$. Since $a_n \to a$, there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < \frac{|a|}{2}, \qquad \forall n \geq N.$$

By the triangle inequality,

$$|a| = |a - a_n + a_n| \leq |a - a_n| + |a_n| < \frac{|a|}{2} + |a_n|.$$

Hence, $|a_n| > \frac{|a|}{2}$, $\forall n \geq N$. This completes the proof. □

Theorem 2.14. *Let (a_n) be a sequence of real numbers such that*

$$\lim_{n\to\infty} a_n = a.$$

If $a_n \geq 0$ for all $n \in \mathbb{N}$, then $a \geq 0$.

Proof. Suppose, on the contrary, that $a < 0$. Then there exists $\delta > 0$ such that $a + \delta < 0$. Since $a_n \to a$, there exists $N \in \mathbb{N}$ such that

$$a_n \in (a - \delta, a + \delta), \qquad \forall n \geq N.$$

Since $a + \delta < 0$, every point in the interval $(a - \delta, a + \delta)$ is negative. Thus,

$$a_n < 0, \qquad \forall n \geq N,$$

which contradicts the assumption that $a_n \geq 0$ for all $n \in \mathbb{N}$. Therefore, $a \geq 0$. □

If convergent sequences are combined using the usual algebraic operations, does the resulting sequence remain convergent? In this section, we show that convergence is preserved under basic algebraic operations.

Theorem 2.15 (Algebra of Convergent Sequences)**.** *Let (a_n) and (b_n) be sequences of real numbers such that $\lim\limits_{n\to\infty} a_n = x$, $\lim\limits_{n\to\infty} b_n = y$, and let $\alpha \in \mathbb{R}$. Then:*

1. $\lim\limits_{n\to\infty} (a_n + b_n) = a + b$.

2. $\lim\limits_{n\to\infty} (\alpha a_n) = \alpha a$.

3. $\lim\limits_{n\to\infty} (a_n b_n) = ab$.

4. *If $a_n > 0$ for all $n \in \mathbb{N}$ and $a > 0$, then* $\lim\limits_{n\to\infty} \frac{1}{a_n} = \frac{1}{a}$.

Proof. Let $\epsilon > 0$ be given.

(1) Since $a_n \to a$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\epsilon}{2}, \qquad \forall n \geq N_1.$$

Similarly, since $b_n \to b$, there exists $N_2 \in \mathbb{N}$ such that

$$|b_n - b| < \frac{\epsilon}{2}, \qquad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence,

$$\lim_{n\to\infty} (a_n + b_n) = a + b.$$

(2) If $\alpha = 0$, then the result is obvious. Assume $\alpha \neq 0$. Since $a_n \to a$, there exists $N \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\epsilon}{|\alpha|}, \qquad \forall n \geq N.$$

Hence,

$$|\alpha a_n - \alpha a| = |\alpha||a_n - a| < \epsilon.$$

Therefore, $\lim\limits_{n\to\infty} (\alpha a_n) = \alpha a$.

(3) Since $a_n \to a$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n| - |a| \leq ||a_n| - |a|| \leq |a_n - a| < 1, \qquad \forall n \geq N_1.$$

Also, there exist $N_2, N_3 \in \mathbb{N}$ such that

$$|a_n - a| < \frac{\epsilon}{2(1 + |b|)}$$

for all $n \geq N_2$, and

$$|b_n - b| < \frac{\epsilon}{2(1 + |a|)}$$

for all $n \geq N_3$. Let $N = \max\{N_1, N_2, N_3\}$. Then

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - a_n b + a_n b - ab| \\ &\leq |a_n||b_n - b| + |b||a_n - a| \\ &\leq (|a| + 1)\frac{\epsilon}{2(1 + |a|)} + |b|\frac{\epsilon}{2(1 + |b|)} < \epsilon. \end{aligned}$$

Hence,

$$\lim_{n\to\infty} (a_n b_n) = ab.$$

(4) Since $a > 0$, there exists $N_1 \in \mathbb{N}$ such that

$$a_n > \frac{a}{2}, \qquad \forall n \geq N_1.$$

Also, there exists $N_2 \in \mathbb{N}$ such that

$$|a_n - a| < \frac{a^2 \epsilon}{2}, \qquad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\left|\frac{1}{a_n} - \frac{1}{a}\right| = \frac{|a_n - a|}{a a_n} \leq \frac{2}{a^2}|a_n - a| < \epsilon.$$

Therefore, $\lim\limits_{n\to\infty} \frac{1}{a_n} = \frac{1}{a}$. $\square$

Exercise 2.1. *Using the ϵ–N definition of the limit of a sequence, prove the following:*

1. *Let (a_n) be the constant sequence defined by $a_n = c$, $\forall n \in \mathbb{N}$, where $c \in \mathbb{R}$ is a constant. Show that*
$$\lim_{n\to\infty} a_n = c.$$
2. *Let $k \in \mathbb{N}$ be fixed and define $a_n = \frac{1}{n^{1/k}}$, $n \in \mathbb{N}$. Show that*
$$\lim_{n\to\infty} \frac{1}{n^{1/k}} = 0.$$
3. *Let $\{a_n\}$ be a bounded sequence and let*
$$\lim_{n\to\infty} b_n = 0.$$
Prove that
$$\lim_{n\to\infty} a_n b_n = 0.$$
Use this result to prove that:
   (a) $\lim_{n\to\infty} \dfrac{\sin n}{n} = 0.$
   (b) $\lim_{n\to\infty} \dfrac{(-1)^n n}{n^2+1} = 0.$
   (c) $\lim_{n\to\infty} (-1)^n a_n = 0$, *if* $\lim_{n\to\infty} a_n = 0.$
4. *Let $\{a_n\}$ and $\{b_n\}$ be two real sequences with*
$$\lim_{n\to\infty} a_n = l, \qquad \lim_{n\to\infty} b_n = m.$$
If
$$x_n = \max\{a_n, b_n\}, \qquad y_n = \min\{a_n, b_n\},$$
prove that the sequences $\{x_n\}$ and $\{y_n\}$ converge to $\max\{l, m\}$ and $\min\{l, m\}$, respectively.

2.3 Monotone sequences

Definition 2.5. 1. *A sequence (a_n) is said to be* ***increasing*** *if*
$$a_n \le a_{n+1}, \qquad \forall n \in \mathbb{N}.$$
2. *A sequence (a_n) is said to be* ***strictly increasing*** *if*
$$a_n < a_{n+1}, \qquad \forall n \in \mathbb{N}.$$
3. *A sequence (a_n) is said to be* ***decreasing*** *if*
$$a_n \ge a_{n+1}, \qquad \forall n \in \mathbb{N}.$$

*4. A sequence (a_n) is said to be **strictly decreasing** if*

$$a_n > a_{n+1}, \qquad \forall n \in \mathbb{N}.$$

*5. A sequence is said to be **monotone** if it is either increasing or decreasing.*

Remark 2.16. *Every increasing sequence is bounded below by x_1. Hence, an increasing sequence is bounded if and only if it is bounded above. Similarly, every decreasing sequence is bounded above by x_1. Therefore, a decreasing sequence is bounded if and only if it is bounded below.*

Example 2.17. *The following are examples of monotone sequences.*

1. *The constant sequence (c) is both increasing and decreasing.*
2. *$\left(\frac{1}{n}\right)$ is strictly decreasing.*
3. *(n) is strictly increasing.*
4. *(2^n) is strictly increasing.*
5. *$\left(\frac{n^2+1}{n^2}\right) = \left(1 + \frac{1}{n^2}\right)$ is strictly decreasing.*
6. *$((-1)^n)$ is not monotone.*

Theorem 2.18 (**Monotone Convergence Theorem**). *1. A monotonically increasing sequence of real numbers is convergent if it is bounded above.*

2. A monotonically decreasing sequence of real numbers is convergent if it is bounded below.

Proof. 1. Let (a_n) be an monotonically increasing sequence that is bounded above. We show that (a_n) is convergent.

Let
$$E = \{a_n : n \in \mathbb{N}\}.$$

Since E is nonempty and bounded above, by the completeness property of $\mathbb{R}$, the set E has a least upper bound. Let
$$\ell = \sup E.$$

Let $\epsilon > 0$ be given. Since $\ell - \epsilon$ is not an upper bound of E, there exists $N \in \mathbb{N}$ such that

$$a_N > \ell - \epsilon.$$

As (a_n) is onotonically increasing, for every $n \geq N$,

$$\ell - \epsilon < a_N \leq a_n \leq \ell < \ell + \epsilon.$$

Hence,
$$|a_n - \ell| < \epsilon, \qquad \forall n \geq N.$$

Therefore,
$$\lim_{n \to \infty} a_n = \ell.$$

2. Left as an exercise.

$\square$

Example 2.19. *1. Let $0 \le r < 1$. Then*

$$\lim_{n\to\infty} r^n = 0.$$

If $r = 0$, the result is immediate. Assume $0 < r < 1$. Then (r^n) is decreasing and bounded below by 0. Hence, by the Monotone Convergence Theorem, (r^n) converges. Let $\lim\limits_{n\to\infty} r^n = \ell$. Then, by the Algebra of Convergent Sequences,

$$\ell = \lim_{n\to\infty} r^{n+1} = r \lim_{n\to\infty} r^n = r\ell.$$

Hence,

$$(1-r)\ell = 0.$$

Since $0 < r < 1$, we have $1 - r > 0$, and therefore $\ell = 0$. Thus, $\lim\limits_{n\to\infty} r^n = 0$.

2. Let $0 \le r < 1$ and define $s_n = \sum\limits_{k=0}^{n} r^k$. Then

$$\lim_{n\to\infty} s_n = \frac{1}{1-r}.$$

If $r = 0$, then $s_n = 1$ for all n. Assume $0 < r < 1$. Since

$$s_{n+1} - s_n = r^{n+1} > 0,$$

the sequence (s_n) is increasing. Moreover,

$$s_n = \frac{1 - r^{n+1}}{1-r} \le \frac{1}{1-r},$$

so (s_n) is bounded above. Hence, (s_n) converges by the Monotone Convergence Theorem. Finally, since as $n \to \infty$, $r^{n+1} \to 0$, and, $s_n = \frac{1-r^{n+1}}{1-r}$, hence

$$\lim_{n\to\infty} s_n = \frac{1}{1-r}.$$

3. Let a sequence (a_n) be defined as follows:

$$a_1 = 2, \qquad a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right), \quad n = 1, 2, \ldots$$

It is seen that, if the sequence converges, then its limit would be $\sqrt{2}$. Since $a_1 = 2$, one may try to show that (a_n) is monotonically decreasing and bounded below. Note that

$$a_{n+1} := \frac{1}{2}\left(a_n + \frac{2}{a_n}\right) \le a_n \iff a_n^2 \ge 2, \quad \text{i.e., } a_n \ge \sqrt{2}.$$

Clearly $a_1 \ge \sqrt{2}$. Now,

$$a_{n+1} := \frac{1}{2}\left(a_n + \frac{2}{a_n}\right) \ge \sqrt{2} \iff a_n^2 - 2\sqrt{2}a_n + 2 \ge 0,$$

i.e., if and only if $(a_n - \sqrt{2})^2 \ge 0$. This is true for every $n \in \mathbb{N}$. Thus, (a_n) is monotonically decreasing and bounded below, so that by Theorem 1.5, (a_n) converges.

4. Define

$$a_n = \left(1+\frac{1}{n}\right)^n, \qquad n \in \mathbb{N}.$$

We show that (a_n) is increasing and bounded above. By the Arithmetic Mean–Geometric Mean Inequality,

$$\frac{n\left(1+\frac{1}{n}\right)+1}{n+1} > \left[\left(1+\frac{1}{n}\right)^n\right]^{\frac{1}{n+1}},$$

that is,

$$1+\frac{1}{n+1} > \left(1+\frac{1}{n}\right)^{\frac{n}{n+1}}.$$

Hence,

$$\left(1+\frac{1}{n+1}\right)^{n+1} > \left(1+\frac{1}{n}\right)^n,$$

so that $u_{n+1} > a_n$ for all $n \in \mathbb{N}$. Using the binomial theorem,

$$a_n = 1+1+\frac{n(n-1)}{2!}\frac{1}{n^2}+\cdots+\frac{n!}{r!(n-r)!}\frac{1}{n^r}+\cdots+\frac{1}{n^n} < 1+1+\frac{1}{2!}+\cdots+\frac{1}{n!}.$$

Since $n! > 2^{n-1}$ for $n \geq 2$,

$$1+\frac{1}{2!}+\cdots+\frac{1}{n!} < 1+\frac{1}{2}+\frac{1}{2^2}+\cdots+\frac{1}{2^{n-1}} < 2.$$

Therefore, $a_n < 3$, $n \in \mathbb{N}$. Thus (a_n) is increasing and bounded above. Hence, by the Monotone Convergence Theorem, (a_n) converges. Its limit is denoted by

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n.$$

Since $u_1 = 2$, we also have

$$2 < a_n < 3, \qquad n \geq 2.$$

5. Define

$$a_n = \sum_{k=0}^{n}\frac{1}{k!} = 1+\frac{1}{1}+\frac{1}{2!}+\cdots+\frac{1}{n!}.$$

We show that $\lim\limits_{n\to\infty} a_n = e$. Since

$$x_{n+1} - a_n = \frac{1}{(n+1)!} > 0,$$

the sequence (a_n) is increasing. Also,

$$a_n < 1+1+\frac{1}{2}+\frac{1}{2^2}+\cdots+\frac{1}{2^{n-1}} < 3,$$

because $n! > 2^{n-1}$ for $n \geq 3$. Hence (a_n) is bounded above and therefore convergent. From the binomial expansion,

$$\left(1+\frac{1}{n}\right)^n < 1+1+\frac{1}{2!}+\cdots+\frac{1}{n!} = a_n,$$

so

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n \leq \lim_{n\to\infty} a_n.$$

Now fix $m \in \mathbb{N}$. For $n > m$,

$$\left(1+\frac{1}{n}\right)^n > 1+1+\frac{1}{2!}\left(1-\frac{1}{n}\right)+\cdots+\frac{1}{m!}\prod_{j=1}^{m-1}\left(1-\frac{j}{n}\right).$$

Letting $n \to \infty$,

$$e \geq 1+1+\frac{1}{2!}+\cdots+\frac{1}{m!} = x_m.$$

Since this holds for every m, $e \geq \lim_{m\to\infty} x_m$. Hence,

$$\boxed{\lim_{n\to\infty} a_n = e.}$$

2.4 Sandwich theorem

Often, the limit of a sequence can be guessed, but a direct proof of convergence is difficult. In such cases, we compare the sequence with two convergent sequences having the same limit. If the given sequence lies between them, then it must converge to the same limit. This is the content of the *Sandwich Theorem.*

Theorem 2.20 (Sandwich Theorem)**.** *Let (a_n), (b_n), and (c_n) be sequences of real numbers such that*

1. $\lim_{n\to\infty} a_n = \alpha$ *and* $\lim_{n\to\infty} b_n = \alpha$*;*
2. $a_n \leq c_n \leq b_n$ *for all* $n \in \mathbb{N}$*.*

Then

$$\lim_{n\to\infty} c_n = \alpha.$$

Proof. Let $\epsilon > 0$ be given. Since $a_n \to \alpha$, there exists $N_1 \in \mathbb{N}$ such that $a_n \in (\alpha-\epsilon, \alpha+\epsilon)$ for all $n \geq N_1$. Similarly, there exists $N_2 \in \mathbb{N}$ such that $b_n \in (\alpha-\epsilon, \alpha+\epsilon)$ for all $n \geq N_2$. Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\alpha-\epsilon < a_n \leq c_n \leq b_n < \alpha+\epsilon.$$

Hence,

$$|c_n - \alpha| < \epsilon, \qquad \forall n \geq N.$$

Therefore,

$$\lim_{n\to\infty} c_n = \alpha.$$

□

Example 2.21. *Define the sequence (a_n) by*

$$a_n = \frac{\sin n}{n}, \qquad n \in \mathbb{N}.$$

We show that $\lim_{n\to\infty} a_n = 0$. Since

$$-1 \leq \sin n \leq 1,$$

we have

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n}, \qquad \forall n \in \mathbb{N}.$$

We already know that

$$\lim_{n\to\infty}\frac{1}{n} = 0 \quad \text{and} \quad \lim_{n\to\infty}\left(-\frac{1}{n}\right) = 0.$$

Hence, by the Sandwich Theorem,

$$\lim_{n\to\infty}\frac{\sin n}{n} = 0.$$

Example 2.22. *1. Show that*

$$\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+\cdots+\frac{1}{\sqrt{n^2+n}}\right)=1.$$

2. Let $a>0$. We show that

$$\lim_{n\to\infty} a^{1/n}=1.$$

3. Let $0<a<b$. We show that

$$\lim_{n\to\infty}(a^n+b^n)^{1/n}=b.$$

Proof. 1. Let $a_n := \sum_{k=1}^{n}\frac{1}{\sqrt{n^2+k}}$. Since $\frac{1}{\sqrt{n^2+n}}\le\frac{1}{\sqrt{n^2+k}}\le\frac{1}{\sqrt{n^2+1}}$ for every $1\le k\le n$, we obtain $u_n\le a_n\le v_n$, where

$$u_n := \frac{n}{\sqrt{n^2+n}} \quad\text{and}\quad v_n := \frac{n}{\sqrt{n^2+1}}.$$

Now, $u_n=\frac{1}{\sqrt{1+\frac{1}{n}}}\to 1$, and $v_n=\frac{1}{\sqrt{1+\frac{1}{n^2}}}\to 1$. Hence, by the Sandwich Theorem,

$$\lim_{n\to\infty} a_n=1.$$

2. We consider three cases.
Case 1. $a=1$. Then $a^{1/n}=1$ for all n, so the result is immediate.
Case 2. $a>1$. Write

$$a^{1/n}=1+x_n,\qquad x_n>0.$$

By Bernoulli's inequality, $a=(1+x_n)^n\ge 1+nx_n$, so

$$0<a^{1/n}-1=x_n\le\frac{a-1}{n}.$$

Hence, given $\epsilon>0$, choose $N>\frac{a-1}{\epsilon}$. Then, for $n\ge N$,

$$|a^{1/n}-1|<\epsilon.$$

Therefore, $\lim_{n\to\infty} a^{1/n}=1$.
Case 3. $0<a<1$. Let $b=\dfrac{1}{a}>1$. By Case 2,

$$\lim_{n\to\infty} b^{1/n}=1.$$

Since $a^{1/n}=\frac{1}{b^{1/n}}$, it follows that $\lim_{n\to\infty} a^{1/n}=1$. Thus,

$$\boxed{\lim_{n\to\infty} a^{1/n}=1,\qquad a>0.}$$

3. Since $a^n>0$, we have $b^n\le a^n+b^n\le 2b^n$. Taking the n-th root, we obtain

$$b\le(a^n+b^n)^{1/n}\le 2^{1/n}b.$$

We already know that $\lim_{n\to\infty}2^{1/n}=1$. Hence, $\lim_{n\to\infty}2^{1/n}b=b$. Therefore, by the Sandwich Theorem,

$$\lim_{n\to\infty}(a^n+b^n)^{1/n}=b.$$

□

Exercise 2.2. *1. Solve the following exercises.*

(a) *Show that* $\lim_{n\to\infty} n^{1/n} = 1$.

(b) *Let* $a > 1$. *Show that* $\lim_{n\to\infty} \frac{n}{a^n} = 0$.

(c) *Let* $a \in \mathbb{R}$. *Show that* $\lim_{n\to\infty} \frac{a^n}{n!} = 0$.

(d) *Let* (x_n) *be a sequence converging to 0. Define* $s_n := \frac{x_1+x_2+\cdots+x_n}{n}$. *Show that* (s_n) *also converges to 0.*

(e) *Let* $x_n \to x$ *and* $y_n \to y$. *Show that*

$$\frac{x_1y_n + x_2y_{n-1} + \cdots + x_ny_1}{n} \longrightarrow xy.$$

2. *Let a sequence* (a_n) *be defined recursively as follows:*

(a)
$$a_1 = \sqrt{3}, \qquad a_{n+1} = \sqrt{3a_n}, \quad n \geq 1.$$

Show that (a_n) *converges and find its limit.*

(b)
$$a_1 = \sqrt{6}, \qquad a_{n+1} = \sqrt{6 + a_n}, \quad n \geq 1.$$

Show that (a_n) *converges and find its limit.*

3. *Let a sequence* (a_n) *be defined by*

$$a_1 > 0, \qquad a_{n+1} = \sqrt{6 + a_n}, \quad n \geq 1.$$

Show that:

(a) (a_n) *is monotonically increasing if* $0 < a_1 < 3$.

(b) (a_n) *is monotonically decreasing if* $a_1 > 3$.

(c) *Find* $\lim_{n\to\infty} a_n$.

4. *Let*

$$x_n := 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

(a) *Show that* (x_n) *is monotonically increasing.*

(b) *Show that* (x_n) *is bounded above.*

(c) *Deduce that* (x_n) *is convergent.*

(d) *Show that*

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e.$$

5.

6. *Evaluate the following limits:*

(a)

$$\lim_{n\to\infty}\left(\frac{1}{(n+1)^2}+\frac{1}{(n+2)^2}+\cdots+\frac{1}{(n+n)^2}\right).$$

(b)

$$\lim_{n\to\infty}\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots 2n}.$$

2.5 Sequences diverging to $\pm\infty$

Definition 2.6. *Let (a_n) be a real sequence. We say that (a_n)* ***diverges to*** $+\infty$ *(or simply diverges to ∞) if, for every $R\in\mathbb{R}$, there exists $N\in\mathbb{N}$ such that*

$$n\geq N\Rightarrow a_n>R.$$

In this case, we write

$$\lim_{n\to\infty}a_n=+\infty.$$

Definition 2.7. *Let (a_n) be a real sequence. We say that (a_n)* ***diverges to*** $-\infty$ *if, for every $R\in\mathbb{R}$, there exists $N\in\mathbb{N}$ such that*

$$n\geq N\Rightarrow a_n<R.$$

In this case, we write

$$\lim_{n\to\infty}a_n=-\infty.$$

Example 2.23.

1. *The sequence (n^2) diverges to $+\infty$, whereas the sequence $(-n)$ diverges to $-\infty$. That is,*

$$\lim_{n\to\infty}n^2=+\infty\quad\text{and}\quad\lim_{n\to\infty}(-n)=-\infty.$$

2. *Define a sequence (a_n) by*

$$a_{2k-1}=1,\qquad a_{2k}=2k,\qquad k\in\mathbb{N}.$$

The sequence (a_n) is unbounded but does not diverge to $+\infty$. Indeed, let $R=1$. For any $N\in\mathbb{N}$, choose $n=2N-1$. Then $n\geq N$, but

$$a_n=a_{2N-1}=1\not>R.$$

Hence (a_n) does not diverge to $+\infty$.

3. *Let $a>1$. Then*

$$\lim_{n\to\infty}a^n=+\infty.$$

Indeed, write $a=1+h$ for some $h>0$. By Bernoulli's inequality,

$$a^n=(1+h)^n\geq 1+nh>nh.$$

Let $R\in\mathbb{R}$ be given. By the Archimedean property, there exists $N\in\mathbb{N}$ such that

$$Nh>R.$$

Then, for every $n \geq N$,

$$a^n > nh \geq Nh > R.$$

Hence $a^n \to +\infty$, as $n \to \infty$.

4. *The sequence $\left((n!)^{1/n}\right)$ diverges to $+\infty$. Let $R > 0$ be given. Since*

$$\lim_{n\to\infty} \frac{R^n}{n!} = 0,$$

there exists $N \in \mathbb{N}$ such that

$$\frac{R^n}{n!} < 1, \qquad n \geq N.$$

Thus $R^n < n!$, and therefore $R < (n!)^{1/n}$, $n \geq N$. Hence $(n!)^{1/n} \longrightarrow +\infty$.

5. *Consider the sequence*

$$a_n = (-1)^n.$$

This sequence is divergent. Moreover, it diverges neither to $+\infty$ nor to $-\infty$. Indeed, the sequence is bounded since $-1 \leq a_n \leq 1$ for all n. Hence, it cannot diverge to $+\infty$ or to $-\infty$. On the other hand, we have already shown that (a_n) is not convergent. Therefore, (a_n) is divergent, but not divergent to $\pm\infty$.

6. *Define*

$$a_n := \sum_{k=1}^{n} \frac{1}{k}, \qquad n \in \mathbb{N}.$$

We show that (a_n) diverges to $+\infty$. Let $R \in \mathbb{R}$ be given. By the Archimedean Property, choose $m \in \mathbb{N}$ such that

$$\frac{m}{2} > R.$$

Now let $N := 2^m$. For every $n \geq N$, we have

$$a_n \geq a_{2^m}.$$

Moreover,

$$a_{2^m} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{m-1}+1} + \cdots + \frac{1}{2^m}\right).$$

Each group after the first two contains at least

$$2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2},$$

where $j = 1, 2, \ldots, m$. Hence $a_{2^m} \geq 1 + \frac{m}{2} > R$. Thus, $a_n \geq x_{2^m} > R$, $n \geq N$. This proves that

$$\lim_{n\to\infty} a_n = +\infty.$$

2.6 Ratio and root tests

Theorem 2.24 (Ratio Test)**.** *Let (a_n) be a sequence of positive real numbers such that*

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \ell.$$

Then the following holds.

1. If $0 \leq \ell < 1$, then

$$\lim_{n\to\infty} a_n = 0.$$

2. If $\ell > 1$, then

$$\lim_{n\to\infty} a_n = +\infty.$$

Remark 2.25. *If $\ell = 1$, no conclusion can be drawn. For example, $a_n = n$ satisfies*

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{n+1}{n} = 1,$$

but (a_n) diverges to $+\infty$. On the other hand,

$$a_n = \frac{n+1}{n}$$

also satisfies $\lim\limits_{n\to\infty} \frac{a_{n+1}}{a_n} = 1$, but $\lim\limits_{n\to\infty} a_n = 1$.

Theorem 2.26 (Root Test)**.** *Let (a_n) be a sequence of positive real numbers such that*

$$\lim_{n\to\infty} \sqrt[n]{a_n} = \ell.$$

Then the following holds.

1. If $0 \leq \ell < 1$, then

$$\lim_{n\to\infty} a_n = 0.$$

2. If $\ell > 1$, then

$$\lim_{n\to\infty} a_n = +\infty.$$

Remark 2.27. *If $\ell = 1$, no conclusion can be drawn. For example, $a_n = n$ satisfies*

$$\lim_{n\to\infty} \sqrt[n]{a_n} = 1,$$

but (a_n) diverges to $+\infty$. On the other hand, $a_n = \frac{1}{n}$ also satisfies

$$\lim_{n\to\infty} \sqrt[n]{a_n} = 1,$$

but $\lim\limits_{n\to\infty} a_n = 0$.

Example 2.28. *Show that*

$$\lim_{n\to\infty} \frac{2^n}{n!} = 0.$$

Indeed,

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{2^{n+1}/(n+1)!}{2^n/n!} = \lim_{n\to\infty} \frac{2}{n+1} = 0.$$

Hence, $\lim\limits_{n\to\infty} u_n = 0$.

Example 2.29. *Show that*

$$\lim_{n\to\infty} \left(\frac{1}{2}\right)^n = 0.$$

Indeed,

$$\lim_{n\to\infty} \sqrt[n]{\left(\frac{1}{2}\right)^n} = \frac{1}{2} < 1.$$

Hence, $\lim\limits_{n\to\infty} u_n = 0$.

Exercise 2.3. *Find the limits of the following sequences* $\{a_n\}$*, where*

1. $a_n = \left(\frac{2n}{2n-1}\right)^{\sqrt{n}}$
2. $a_n = \left(1 + \frac{1}{n+1}\right)^{1+\frac{1}{n}}$
3. $a_n = \left(\frac{n}{n+2}\right)^{n}$
4. $a_n = \left(\frac{n+1}{n+3}\right)^{n}$

2.7 Subsequences

Definition 2.8. *Let* (r_n) *be a strictly increasing sequence of natural numbers. If* (a_n) *is any sequence, then the sequence*

$$(a_{r_n})$$

is called a ***subsequence*** *of* (a_n)*.*

One can view a subsequence of (a_n) as an infinite part of the sequence obtained by deleting some terms, while keeping the remaining terms in the same order.

Example 2.30. *Let*

$$a_n = 2^n, \qquad n \in \mathbb{N}.$$

Then

$$\{2^{2n}\}_{n\in\mathbb{N}} = \{4, 16, 64, \ldots\}$$

is a subsequence of (a_n)*. Here, the corresponding indices are*

$$r_n = 2n, \qquad n \in \mathbb{N},$$

which form a strictly increasing sequence.

Remark 2.31. *A subsequence of* (a_n) *can also be written as* $(a_{f(n)})$*, where*

$$f : \mathbb{N} \to \mathbb{N}$$

is a strictly increasing function. Here, the n*th term of the subsequence is the* $f(n)$*th term of the original sequence.*

Example 2.32. *Let* (a_n) *be any sequence.*

1. *The sequence*

   $$(a_2, a_4, a_6, \ldots)$$

   is called the even subsequence of (a_n)*. Here,*

   $$r_n = 2n, \qquad n \in \mathbb{N}.$$

2. *The sequence*

   $$(a_1, a_3, a_5, \ldots)$$

   is called the odd subsequence of (a_n)*. Here,*

   $$r_n = 2n - 1, \qquad n \in \mathbb{N}.$$

Example 2.33. *The following are subsequences of the given sequences.*

1. *Let* $a_n = \dfrac{1}{n}$ *and* $r_n = 2n$. *Then*

$$(a_{r_n}) = (a_{2n}) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \ldots\right)$$

   is a subsequence of $\left(\dfrac{1}{n}\right)$.

2. *Let* $a_n = \dfrac{1}{n}$ *and* $r_n = 2n-1$. *Then*

$$(a_{r_n}) = (a_{2n-1}) = \left(1, \frac{1}{3}, \frac{1}{5}, \ldots\right)$$

   is a subsequence of $\left(\dfrac{1}{n}\right)$.

3. *Let* $a_n = (-1)^n$ *and* $r_n = 2n$. *Then*

$$(a_{r_n}) = (a_{2n}) = (1, 1, 1, \ldots)$$

   is a subsequence of $((-1)^n)$.

4. *Let* $a_n = 1 + \dfrac{1}{n}$ *and* $r_n = n^2$. *Then*

$$(a_{r_n}) = (a_{n^2}) = \left(2, 1 + \frac{1}{2^2}, 1 + \frac{1}{3^2}, \ldots\right)$$

   is a subsequence of $\left(1 + \dfrac{1}{n}\right)$.

Theorem 2.34. *Let* (a_n) *be a sequence converging to* x. *If* (a_{r_n}) *is a subsequence of* (a_n), *then*

$$a_{r_n} \to x.$$

Proof. Let $\epsilon > 0$ be given. Since $a_n \to x$, there exists $N \in \mathbb{N}$ such that

$$|a_n - x| < \epsilon, \qquad \forall n \geq N.$$

Since (r_n) is a strictly increasing sequence of natural numbers, we have $r_n \geq n$ for every $n \in \mathbb{N}$. Hence, if $n \geq N$, then

$$r_n \geq n \geq N,$$

and therefore

$$|a_{r_n} - x| < \epsilon.$$

Thus,

$$a_{r_n} \to x.$$

□

Example 2.35. *Show that*

$$\lim_{n\to\infty} \left(1 + \frac{1}{2n}\right)^n = \sqrt{e}.$$

Proof. Let $a_n = \left(1+\frac{1}{n}\right)^n$. We have already shown that $a_n \to e$. Hence, every subsequence of (a_n) also converges to e. In particular,

$$a_{2n} = \left(1+\frac{1}{2n}\right)^{2n} \longrightarrow e.$$

Therefore, $\left[\left(1+\frac{1}{2n}\right)^n\right]^2 \longrightarrow e$. Since $\left(1+\frac{1}{2n}\right)^n > 0$ for every n, it follows that

$$\left(1+\frac{1}{2n}\right)^n \longrightarrow \sqrt{e}.$$

$\square$

Remark 2.36. *The existence of two convergent subsequences with the same limit does not guarantee that the original sequence is convergent. For example, let*

$$a_n = \sin\left(\frac{n\pi}{4}\right).$$

Then

$$(a_{8n-7}) = \left(\sin\frac{\pi}{4}, \sin\frac{9\pi}{4}, \sin\frac{17\pi}{4}, \dots\right) \longrightarrow \frac{1}{\sqrt{2}},$$

and

$$(a_{8n-5}) = \left(\sin\frac{3\pi}{4}, \sin\frac{11\pi}{4}, \sin\frac{19\pi}{4}, \dots\right) \longrightarrow \frac{1}{\sqrt{2}}.$$

However, (a_n) is not convergent.

Theorem 2.37 (Existence of a Monotone Subsequence). *Every real sequence has a monotone subsequence.*

Theorem 2.38 (Bolzano-Weierstrass Theorem). *Every bounded real sequence has a convergent subsequence.*

Proof. Let (a_n) be a bounded sequence. Then there exists a closed and bounded interval

$$I_0 = [a, b]$$

containing every term of the sequence. Bisect I_0. At least one of the two halves contains infinitely many terms of (a_n); denote it by I_1. Repeating this construction, we obtain a sequence of closed intervals (I_n) such that

1. $I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$;
2. $|I_n| = \dfrac{b-a}{2^n}$, so that
$$\lim_{n\to\infty} |I_n| = 0;$$
3. each I_n contains infinitely many terms of (a_n).

By Cantor's Nested Interval Theorem,

$$\bigcap_{n=0}^{\infty} I_n = \{\alpha\}$$

for some $\alpha \in \mathbb{R}$. We show that α is a subsequential limit of (a_n). Let $\epsilon > 0$. Choose $k \in \mathbb{N}$ such that

$$|I_k| < \epsilon.$$

Since $\alpha \in I_k$,

$$I_k \subseteq (\alpha - \epsilon, \alpha + \epsilon).$$

As I_k contains infinitely many terms of (a_n), every ϵ-neighborhood of α contains infinitely many terms of the sequence. Hence, α is a subsequential limit of (a_n). Therefore, there exists a subsequence of (a_n) converging to α. This completes the proof. ☐

Example 2.39. *Show that the sequence $((-1)^n)$ is divergent.*

Proof. Suppose, on the contrary, that $((-1)^n)$ converges to some real number ℓ. Then every subsequence must also converge to ℓ. However, the even subsequence is

$$((-1)^{2n}) = (1, 1, 1, \ldots),$$

which converges to 1, whereas the odd subsequence is

$$((-1)^{2n-1}) = (-1, -1, -1, \ldots),$$

which converges to -1.

Since a convergent sequence cannot have two subsequences with different limits, we obtain a contradiction. Hence $((-1)^n)$ is divergent. ☐

2.8 Cauchy sequences

So far, to prove that a sequence converges, we first identified its limit and then verified the definition of convergence. This leads to a natural question:

Can we determine whether a sequence is convergent without knowing its limit?

If (a_n) converges, then for sufficiently large m, n, the terms a_m and a_n are arbitrarily close to each other. Indeed, by the triangle inequality,

$$|a_n - a_m| \leq |a_n - x| + |a_m - x|,$$

which can be made arbitrarily small when m and n are sufficiently large. This observation leads to the following definition.

Definition 2.9 (Cauchy Sequence). *A sequence (a_n) of real numbers is said to be **Cauchy** if, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$|a_n - a_m| < \epsilon, \qquad \forall m, n \geq N.$$

Example 2.40. *Every convergent sequence is a Cauchy sequence. Is the converse true?*

Theorem 2.41 (Cauchy Criterion). *A sequence of real numbers is Cauchy if and only if it is convergent.*

Proof. We have already seen that every convergent sequence is Cauchy. We now prove the converse.

Let $E = \{x \in \mathbb{R} : \exists N \in \mathbb{N} \text{ such that } x < a_n \ \forall n \geq N\}$.

Let $\delta > 0$. Since (a_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|a_m - a_n| < \delta$ for all $m, n \geq N_0$. In particular, $|a_{N_0} - a_n| < \delta$ for all $n \geq N_0$. Hence, $a_n \in (a_{N_0} - \delta, a_{N_0} + \delta)$ for all $n \geq N_0$. In particular, $a_n < a_{N_0} + \delta$ for all $n \geq N_0$.

Claim 1. $a_{N_0} - \delta \in E$. Indeed, taking $N = N_0$, we have $a_n > a_{N_0} - \delta$ for all $n \geq N_0$.

Claim 2. $a_{N_0}+\delta$ is an upper bound of E. Suppose not. Then there exists $x \in E$ such that $x > a_{N_0}+\delta$. Since $x \in E$, there exists $N \in \mathbb{N}$ such that $a_n > x$ for all $n \geq N$. Hence, for all $n \geq \max\{N, N_0\}$,

$$a_n > x > a_{N_0}+\delta,$$

which contradicts the inequality $a_n < a_{N_0}+\delta$ for all $n \geq N_0$.

Thus, E is nonempty and bounded above. By the completeness of $\mathbb{R}$, let $\ell = \sup E$.

Claim 3. $a_n \to \ell$. Let $\epsilon > 0$. Since (a_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|a_n - a_{N_0}| < \epsilon/2$ for all $n \geq N_0$. By Claim 1, $a_{N_0} - \epsilon/2 \in E$, and hence $a_{N_0} - \epsilon/2 \leq \ell$. By Claim 2, $\ell \leq a_{N_0} + \epsilon/2$. Therefore, $|a_{N_0} - \ell| \leq \epsilon/2$. Now, for every $n \geq N_0$,

$$|a_n - \ell| \leq |a_n - a_{N_0}| + |a_{N_0} - \ell| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence, $a_n \to \ell$. □

Example 2.42. *1. The sequence $\left(\frac{1}{n}\right)$ is a Cauchy sequence.*

2. The sequence $((-1)^n)$ is not a Cauchy sequence.

3. Let

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}, \qquad n \in \mathbb{N}.$$

We show, using the Cauchy Criterion, that (s_n) is not convergent.

Proof. 1. Let $\epsilon > 0$ be given. Choose $N \in \mathbb{N}$ such that $N > \frac{1}{\epsilon}$. Then, for any $m, n \geq N$ without loss of generality, assume $m \leq n$. Hence,

$$\left|\frac{1}{m} - \frac{1}{n}\right| = \frac{1}{m} - \frac{1}{n} < \frac{1}{m} \leq \frac{1}{N} < \epsilon.$$

Therefore, $\left(\frac{1}{n}\right)$ is a Cauchy sequence.

2. Choose $\epsilon = 1$. Let $N \in \mathbb{N}$ be arbitrary. Take

$$m = 2N, \qquad n = 2N+1.$$

Then $m, n \geq N$, $(-1)^m = 1$, and $(-1)^n = -1$. Hence,

$$|(-1)^m - (-1)^n| = |1 - (-1)| = 2 > 1 = \epsilon.$$

Therefore, $((-1)^n)$ is not a Cauchy sequence.

3. By the Cauchy Criterion, it is enough to show that (s_n) is not a Cauchy sequence. Choose $\epsilon = \frac{1}{2}$ and let $N \in \mathbb{N}$ be arbitrary. Take $n \geq N$ and let $m = 2n$. Then

$$\begin{aligned} |s_m - s_n| &= |s_{2n} - s_n| \\ &= \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \\ &\geq \frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n} = \frac{1}{2}. \end{aligned}$$

Thus, for every $N \in \mathbb{N}$, there exist $m, n \geq N$ such that

$$|s_m - s_n| \geq \frac{1}{2}.$$

Hence, (s_n) is not a Cauchy sequence. Therefore, by the Cauchy Criterion, (s_n) is not convergent.

□

Example 2.43. *Let (a_n) be a sequence of real numbers. Suppose there exists a positive real number $\rho < 1$ such that*

$$|a_{n+1} - a_n| \le \rho |a_n - a_{n-1}| \quad \forall n \in \mathbb{N}, n \ge 2.$$

Then (a_n) is a Cauchy sequence. To see this first we observe that

$$|a_{n+1} - a_n| \le \rho^{n-1} |a_2 - a_1| \quad \forall n \in \mathbb{N}, n \ge 2.$$

Hence, for $n > m$,

$$\begin{aligned} |a_n - a_m| &\le |a_n - a_{n-1}| + \ldots + |a_{m+1} - a_m| \\ &\le (\rho^{n-2} + \ldots + \rho^{m-1}) |a_2 - a_1| \\ &\le \rho^{m-1} (1 + \rho + \ldots + \rho^{n-m-3}) |a_2 - a_1| \\ &\le \frac{\rho^{m-1}}{1-\rho} |a_2 - a_1|. \end{aligned}$$

Since $\rho^{m-1} \to 0$ as $m \to \infty$, given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - a_m| < \epsilon$ for all $n, m \ge N$.

Exercise 2.4.

1. *Let (x_n) be a sequence. Suppose that both the even subsequence (x_{2n}) and the odd subsequence (x_{2n-1}) converge to the same limit x. Show that the original sequence (x_n) also converges to x.*
2. *Apply the Bolzano-Weierstrass Theorem to solve the following problems:*
   (a) *Show that the sequence $(\sin n)$ has a convergent subsequence.*
   (b) *Prove that every bounded sequence has a monotone convergent subsequence.*
   (c) *True or False: A sequence (x_n) is bounded if and only if every subsequence of (x_n) has a convergent subsequence.*
3. *Let (a_n) and (b_n) be Cauchy sequences. Prove directly that:*
   (a) *$(a_n + b_n)$ is a Cauchy sequence.*
   (b) *$(a_n b_n)$ is a Cauchy sequence.*
4. *Let (a_n) be a Cauchy sequence in $\mathbb{R}$ having a subsequence converging to a real number l. Prove that*
   $$\lim_{n \to \infty} a_n = l.$$
5. *Consider the following recursively defined sequences:*
   (a) *Let*
   $$a_1 = 2, \qquad a_{n+1} = 2 + \frac{1}{a_n}, \quad n \ge 1.$$
   Prove that (a_n) converges and find its limit.
   (b) *Let*
   $$a_1 > 0, \qquad a_{n+1} = \frac{1}{2 + a_n}, \quad n \ge 1.$$
   Prove that (a_n) converges and find its limit.

3 Introduction to Infinite Series

In the previous chapter, we studied *sequences*, that is, ordered infinite collections of real numbers. A natural question is:

Can we meaningfully add infinitely many real numbers?

For a finite collection of real numbers $a_1, a_2, \dots, a_n$, the sum

$$a_1 + a_2 + \cdots + a_n$$

is well defined. It is therefore natural to consider the infinite expression

$$\sum_{n=1}^{\infty} a_n.$$

Unlike finite sums, however, infinite sums cannot be manipulated arbitrarily. Without a precise definition, they may lead to contradictions. For example, consider the series

$$1 - 1 + 1 - 1 + \cdots .$$

If we group the terms as

$$(1-1) + (1-1) + (1-1) + \cdots ,$$

we obtain

$$0 + 0 + 0 + \cdots = 0.$$

But if we group them as

$$1 + (-1+1) + (-1+1) + \cdots ,$$

we obtain

$$1 + 0 + 0 + \cdots = 1.$$

To avoid such contradictions, we need a precise definition of the sum of an infinite series. Developing this definition and studying its properties is the main objective of this chapter.

3.1 Convergence and sum of an infinite series

Definition 3.1. *Let (a_n) be a sequence of real numbers. A formal expression of the form*

$$\sum_{n=1}^{\infty} a_n,$$

or simply $\sum a_n$ or $\sum_n a_n$ is called an ***infinite series****.*

Definition 3.2 (Partial Sum)**.** *Let $\sum_{n=1}^{\infty} a_n$ be an infinite series. For each $n \in \mathbb{N}$, the finite sum*

$$s_n := a_1 + a_2 + \cdots + a_n$$

is called the nth ***partial sum*** *of the series.*

Remark 3.1. *Associated with the series* $\sum_{n=1}^{\infty} a_n$*, we have the sequence of partial sums* (s_n)*, where*

$$s_n = a_1 + a_2 + \cdots + a_n, \qquad n \in \mathbb{N}.$$

Thus, every infinite series naturally determines a sequence of partial sums.

Definition 3.3 (Convergent Series)**.** *An infinite series* $\sum_{n=1}^{\infty} a_n$ *is said to be* ***convergent*** *if the sequence of partial sums* (s_n) *is convergent.*

Remark 3.2. *If* $s_n \longrightarrow s$*, then we call* s *the* ***sum*** *of the series and write* $\sum_{n=1}^{\infty} a_n = s$*.*

Example 3.3. 1. *Consider the series* $\frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \cdots$ *. Let* $\sum_{n=1}^{\infty} a_n$*, where* $a_n = \frac{1}{n(n+1)}$*. If* $s_n = u_1 + u_2 + \cdots + a_n$*, then*

$$\begin{aligned} s_n &= \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

Hence, $\lim_{n\to\infty} s_n = 1$*. Therefore, the series* $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ *is convergent and its sum is* 1*.*

2. ***(Geometric Series)****Let* $a \in \mathbb{R}$ *satisfy* $|a| < 1$*. Consider the infinite series* $\sum_{n=0}^{\infty} a^n$*. Since the nth partial sum is*

$$s_n = \sum_{k=0}^{n} a^k = \frac{1 - a^{n+1}}{1 - a},$$

and $a^{n+1} \longrightarrow 0$*, it follows that* $s_n \longrightarrow \frac{1}{1-a}$*. Thus,* $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \qquad |a| < 1.$

Definition 3.4 (**Divergent Series**)**.** *An infinite series* $\sum_{n=1}^{\infty} a_n$ *is said to be* ***divergent*** *if the sequence of partial sums* (s_n) *is divergent.*

Example 3.4. 1. *Consider the series* $1 + 2 + 3 + \cdots$ *. Its sequence of partial sums is*

$$s_n = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Since $\lim_{n\to\infty} s_n = \infty$*, the sequence of partial sums does not converge. Hence, the series* $\sum_{n=1}^{\infty} n$ *is divergent.*

2. ***(Harmonic Series)*** *Consider the series*

$$1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots.$$

Let $\sum\limits_{n=1}^{\infty} a_n$ *be the series, where* $a_n = \frac{1}{n}$. *Let* $s_n = a_1 + a_2 + \cdots + a_n$ *be the sequence of partial sums. Then*

$$\begin{aligned}
s_2 &= 1+\frac{1}{2},\\
s_4 &= 1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4} > 1+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right) = 1+2\cdot\frac{1}{2},\\
s_8 &= 1+\frac{1}{2}+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right) > 1+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right) = 1+3\cdot\frac{1}{2},\\
s_{16} &> 1+4\cdot\frac{1}{2},\\
&\vdots
\end{aligned}$$

and, in general,

$$s_{2^n} > 1+n\cdot\frac{1}{2}.$$

Hence, $\lim\limits_{n\to\infty} s_{2^n} = \infty$. *Since* $s_{n+1} - s_n = a_{n+1} > 0$ *for all* $n \in \mathbb{N}$, *the sequence* $\{s_n\}$ *is monotone increasing. As the subsequence* $\{s_{2^n}\}$ *is unbounded, the sequence* $\{s_n\}$ *is also unbounded. Therefore,* $\sum\limits_{n=1}^{\infty} \frac{1}{n}$ *is divergent.*

3. ***(p-Series)*** *For* $p > 0$, *consider the series* $\sum\limits_{n=1}^{\infty} \frac{1}{n^p}$. *Then*

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is convergent if and only if $p > 1$. *Equivalently,*

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \infty, & 0 < p \le 1, \\ \textit{converges}, & p > 1. \end{cases}$$

In particular, the harmonic series $\sum\limits_{n=1}^{\infty} \frac{1}{n}$ *is divergent, whereas the series* $\sum\limits_{n=1}^{\infty} \frac{1}{n^2}$ *is convergent.*

Theorem 3.5. *Let* m *be a natural number. Then the two series* $\sum\limits_{n=1}^{\infty} a_n$ *and* $\sum\limits_{n=m+1}^{\infty} a_n$ *converge or diverge together.*

Proof. Let $s_n = a_1 + a_2 + \cdots + a_n$, and $t_n = a_{m+1} + a_{m+2} + \cdots + a_{m+n}$. Then $t_n = s_{m+n} - s_m$, where s_m is a fixed number. If $\{s_n\}$ converges, then $\{t_n\}$ converges, and conversely. If $\{s_n\}$ diverges, then $\{t_n\}$ diverges, and conversely. Hence, both the sequences $\{s_n\}$ and $\{t_n\}$, and consequently the series $\sum\limits_{n=1}^{\infty} a_n$ and $\sum\limits_{n=m+1}^{\infty} a_n$, converge or diverge together. □

Remark 3.6. *The theorem states that removing a finite number of terms from the beginning of a series, or adding a finite number of terms to the beginning of a series, does not affect its convergence or divergence.*

Theorem 3.7 (**Algebra of Convergent Series**). *Let* $\sum_{n=1}^{\infty} a_n$ *and* $\sum_{n=1}^{\infty} b_n$ *be two convergent series with sums* A *and* B*, respectively. Then:*

1. *The series* $\sum_{n=1}^{\infty} (a_n + b_n)$ *is convergent and*

$$\sum_{n=1}^{\infty} (a_n + b_n) = A + B.$$

2. *For every* $\lambda \in \mathbb{R}$*, the series* $\sum_{n=1}^{\infty} \lambda a_n$ *is convergent and*

$$\sum_{n=1}^{\infty} \lambda a_n = \lambda A.$$

Theorem 3.8 (**Cauchy Criterion**). *The series* $\sum_{n=1}^{\infty} a_n$ *converges if and only if for every* $\epsilon > 0$ *there exists* $N \in \mathbb{N}$ *such that*

$$n, m \geq N \Rightarrow |s_n - s_m| < \epsilon,$$

where (s_n) *denotes the sequence of partial sums. Thus, the series* $\sum_{n=1}^{\infty} a_n$ *converges if and only if for every* $\epsilon > 0$ *there exists* $N \in \mathbb{N}$ *such that*

$$n > m \geq N \Rightarrow |a_{m+1} + a_{m+2} + \cdots + a_n| < \epsilon.$$

Proof. The series $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence (s_n) of partial sums converges. By the Cauchy Criterion for sequences, this is equivalent to saying that (s_n) is a Cauchy sequence. Since

$$s_n - s_m = a_{m+1} + a_{m+2} + \cdots + a_n,$$

the second statement follows immediately. □

Theorem 3.9. *Suppose that the series* $\sum_{n=1}^{\infty} a_n$ *converges to* s*. Then, for every* $\epsilon > 0$*, there exists* $N \in \mathbb{N}$ *such that*

$$\left| \sum_{n=N+1}^{\infty} a_n \right| < \epsilon.$$

Proof. Let $\epsilon > 0$ be given. Since $s_n \to s$, there exists $N \in \mathbb{N}$ such that $|s - s_N| < \epsilon$. Since $\sum_{n=N+1}^{\infty} a_n = s - s_N$, it follows that

$$\left| \sum_{n=N+1}^{\infty} a_n \right| = |s - s_N| < \epsilon.$$

This completes the proof. □

Theorem 3.10. *If the series* $\sum_{n=1}^{\infty} a_n$ *converges, then* $\lim_{n \to \infty} a_n = 0$.

Proof. Let $s_n = \sum_{k=1}^{n} a_k$ be the sequence of partial sums. Since the series converges, (s_n) converges. Hence, by the Cauchy Criterion for sequences, for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|s_n - s_m| < \epsilon, \qquad \forall n, m \geq N.$$

Now let $n \geq N + 1$. Then $n - 1 \geq N$, and therefore

$$|a_n| = |s_n - s_{n-1}| < \epsilon.$$

Hence $a_n \to 0$. $\square$

Example 3.11. *Show that the series* $\sum_{n=1}^{\infty} a_n$ *is divergent, where* $a_n = \frac{n}{n+1}$.

Proof. Since $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{n+1} = 1 \neq 0$, the necessary condition for the convergence of a series is violated. Hence $\sum_{n=1}^{\infty} a_n$ is divergent. $\square$

Theorem 3.12. *If the series* $\sum_{n=1}^{\infty} |a_n|$ *converges, then the series* $\sum_{n=1}^{\infty} a_n$ *also converges.*

Proof. Let $s_n = \sum_{k=1}^{n} a_k$ and $\sigma_n = \sum_{k=1}^{n} |a_k|$ denote the sequences of partial sums of $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} |a_n|$, respectively.

Since (σ_n) converges, it is a Cauchy sequence. Thus, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|\sigma_n - \sigma_m| < \epsilon, \qquad \forall n > m \geq N.$$

Then

$$|s_n - s_m| = |a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n| = \sigma_n - \sigma_m < \epsilon.$$

Hence, (s_n) is a Cauchy sequence. By the Cauchy Criterion for sequences, (s_n) converges. Therefore, $\sum_{n=1}^{\infty} a_n$ converges. $\square$

Example 3.13. *Show that the series* $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} a_n$, *where* $a_n = \frac{(-1)^{n+1}}{n}$, *is convergent.*

Proof. Let $s_n = a_1 + a_2 + \cdots + a_n$ be the sequence of partial sums. Then, for any $p \in \mathbb{N}$,

$$|s_{n+p} - s_n| = \left| \frac{1}{n+1} - \frac{1}{n+2} + \frac{1}{n+3} - \cdots + (-1)^{p-1}\frac{1}{n+p} \right| < \frac{1}{n+1}.$$

Let $\varepsilon > 0$. Then $|s_{n+p} - s_n| < \varepsilon$ whenever $n > \frac{1}{\varepsilon} - 1$. Choose $m = \left\lfloor \frac{1}{\varepsilon} - 1 \right\rfloor + 2$. Then m is a natural number and

$$|s_{n+p} - s_n| < \varepsilon$$

for all $n \geq m$ and $p = 1, 2, 3, \ldots$. Hence $\{s_n\}$ is a Cauchy sequence. Since $\mathbb{R}$ is complete, $\{s_n\}$ converges. Therefore, $\sum_{n=1}^{\infty} a_n$ is convergent. $\square$

Exercise 3.1. *1. Consider the constant sequence $a_n = c$, $n \in \mathbb{N}$, where $c \in \mathbb{R}$ is fixed. Determine whether the series $\sum_{n=1}^{\infty} a_n$ is convergent.*

2. Find the sum of each of the following infinite series, if it exists.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n + 2^{n+1}}$.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{3}{(3n-2)(3n+1)}$.

3. Using the Cauchy criterion, determine whether each of the following series converges.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n}$.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$.

3.2 Tests of convergence of a series of positive terms

The convergence or divergence of a series is determined by its sequence of partial sums. Since partial sums are often difficult to compute explicitly, we use *convergence tests*, which determine the nature of a series without evaluating its partial sums directly. We now study some of these tests.

Definition 3.5. *A series $\sum_{n=1}^{\infty} a_n$ is called a* series of positive terms *if $a_n > 0$, for all $n \in \mathbb{N}$.*

Theorem 3.14. *A series of positive real numbers $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the sequence of partial sums $\{s_n\}$, $s_n = \sum_{k=1}^{n} a_k$, is bounded above.*

Remark 3.15. *Since $s_n = a_1 + a_2 + \cdots + a_n$ is a monotone increasing sequence, if it is not bounded above, then $\lim_{n\to\infty} s_n = \infty$. Hence, the series $\sum_{n=1}^{\infty} a_n$ diverges to ∞. Therefore, a series of positive real numbers either converges to a finite real number or diverges to ∞.*

Theorem 3.16 (Comparison Test). *Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series of nonnegative real numbers. Suppose that $a_n \le b_n$ for all $n \in \mathbb{N}$. Then:*

1. *If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ also converges.*
2. *If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ also diverges.*

Example 3.17. *The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent. Since*

$$0 \le \frac{1}{n^3} \le \frac{1}{n^2}, \qquad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum\limits_{n=1}^{\infty} \frac{1}{n^3}$ *is also convergent.*

Example 3.18. *The harmonic series* $\sum\limits_{n=1}^{\infty} \frac{1}{n}$ *is divergent. Since*

$$\frac{1}{n} \le \frac{1}{\sqrt{n}}, \qquad \forall n \in \mathbb{N},$$

the Comparison Test implies that $\sum\limits_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ *is also divergent.*

Theorem 3.19 (Limit Form of the Comparison Test). *Let* $\sum\limits_{n=1}^{\infty} a_n$ *and* $\sum\limits_{n=1}^{\infty} b_n$ *be two series of positive real numbers. If*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = l,$$

where $0 < l < \infty$*, then the two series either both converge or both diverge.*

Example 3.20. *Test the convergence of the series*

$$\frac{1+2}{2^3} + \frac{1+2+3}{3^3} + \frac{1+2+3+4}{4^3} + \cdots .$$

Proof. Let $\sum\limits_{n=1}^{\infty} a_n$ be the given series. Then

$$a_n = \frac{1+2+\cdots+(n+1)}{(n+1)^3} = \frac{(n+1)(n+2)}{2(n+1)^3} = \frac{n+2}{2(n+1)^2}.$$

Choose $b_n = \frac{1}{n}$. Then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n(n+2)}{2(n+1)^2} = \frac{1}{2}.$$

Since $\sum\limits_{n=1}^{\infty} \frac{1}{n}$ diverges, the given series also diverges by the Limit Comparison Test. ☐

Example 3.21. *Test the convergence of the series*

$$\frac{1}{1\cdot 2^2} + \frac{1}{2\cdot 3^2} + \frac{1}{3\cdot 4^2} + \cdots .$$

Proof. Let $\sum\limits_{n=1}^{\infty} a_n$ be the given series. Then $a_n = \frac{1}{n(n+1)^2}$. Choose $b_n = \frac{1}{n^3}$. Then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{n^2}{(n+1)^2} = 1.$$

Since $\sum\limits_{n=1}^{\infty} \frac{1}{n^3}$ converges, the given series also converges by the Limit Comparison Test. ☐

Remark 3.22.

1. *If* $\lim\limits_{n\to\infty} \frac{a_n}{b_n} = 0$*, then* $\sum\limits_{n=1}^{\infty} a_n$ *converges whenever* $\sum\limits_{n=1}^{\infty} b_n$ *converges.*

2. *If* $\lim\limits_{n\to\infty} \frac{a_n}{b_n} = \infty$*, then* $\sum\limits_{n=1}^{\infty} a_n$ *diverges whenever* $\sum\limits_{n=1}^{\infty} b_n$ *diverges.*

Theorem 3.23 (p-Series Test). *The series* $\sum\limits_{n=1}^{\infty} \frac{1}{n^p}$ *converges if* $p > 1$ *and diverges if* $p \le 1$*.*

Theorem 3.24 (**d'Alembert's Ratio Test**). *Let* $\sum_{n=1}^{\infty} a_n$ *be a series of positive real numbers. Suppose that*

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = r.$$

Then:

1. *If* $0 \le r < 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *converges.*
2. *If* $r > 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*
3. *If* $r = 1$, *then the test is inconclusive.*

Example 3.25.

1. *Consider the series* $\sum_{n=1}^{\infty} \frac{1}{n!}$. *Since*

$$\frac{a_{n+1}}{a_n} = \frac{1}{n+1} \longrightarrow 0,$$

the Ratio Test implies that the series converges.

2. *Consider the series* $\sum_{n=1}^{\infty} \frac{2^n}{n}$. *Since*

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{n+1} \cdot \frac{n}{2^n} = 2 \cdot \frac{n}{n+1} \longrightarrow 2 > 1,$$

the Ratio Test implies that the series diverges.

3. *Consider the harmonic series* $\sum_{n=1}^{\infty} \frac{1}{n}$. *Since*

$$\frac{a_{n+1}}{a_n} = \frac{n}{n+1} \longrightarrow 1,$$

the Ratio Test is inconclusive. Indeed, the harmonic series is divergent. On the other hand, for the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$,

$$\frac{a_{n+1}}{a_n} = \left(\frac{n}{n+1}\right)^2 \longrightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Theorem 3.26 (**Cauchy's Root Test**). *Let* $\sum_{n=1}^{\infty} a_n$ *be a series of positive real numbers. Suppose that*

$$\lim_{n\to\infty} \sqrt[n]{a_n} = r.$$

Then:

1. *If* $0 \le r < 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *converges.*
2. *If* $r > 1$, *then the series* $\sum_{n=1}^{\infty} a_n$ *diverges.*
3. *If* $r = 1$, *then the test is inconclusive.*

Example 3.27. 1. *Consider the series* $\sum\limits_{n=1}^{\infty} \frac{1}{2^n}$. *Since*

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{2^n}} = \frac{1}{2},$$

the Root Test implies that the series converges.

2. *Consider the series* $\sum\limits_{n=1}^{\infty} 2^n$. *Since*

$$\sqrt[n]{a_n} = \sqrt[n]{2^n} = 2,$$

the Root Test implies that the series diverges.

3. *Consider the harmonic series* $\sum\limits_{n=1}^{\infty} \frac{1}{n}$. *Since*

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n}} = \frac{1}{n^{1/n}} \longrightarrow 1,$$

the Root Test is inconclusive. Indeed, the harmonic series is divergent. On the other hand, for the series $\sum\limits_{n=1}^{\infty} \frac{1}{n^2}$,

$$\sqrt[n]{a_n} = \sqrt[n]{\frac{1}{n^2}} = \frac{1}{n^{2/n}} \longrightarrow 1,$$

yet this series is convergent. Thus, no conclusion can be drawn when the limit equals 1.

Example 3.28. *Show that the series* $\sum\limits_{n=1}^{\infty} \frac{2^n n!}{n^n}$ *converges.*

Proof. Let $a_n = \frac{2^n n!}{n^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = 2\left(\frac{n}{n+1}\right)^n.$$

Since $\left(1+\frac{1}{n}\right)^n \to e$, we have

$$\left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n} \longrightarrow \frac{1}{e}.$$

Hence

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \frac{2}{e} < 1.$$

Therefore, by the Ratio Test, the series $\sum\limits_{n=1}^{\infty} \frac{2^n n!}{n^n}$ converges. $\square$

Theorem 3.29 (Gauss's Test). *Let* $\sum\limits_{n=1}^{\infty} a_n$ *be a series of positive real numbers. Suppose* $\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^p}$, *where* $p > 1$ *and* $\{b_n\}$ *is a bounded sequence. Then* $\sum\limits_{n=1}^{\infty} a_n$ *converges if* $\alpha > 1$ *and diverges if* $\alpha \leq 1$.

Example 3.30. *Test the convergence of the series*

$$1 + \frac{2^2}{3^2} + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2} + \frac{2^2 \cdot 4^2 \cdot 6^2}{3^2 \cdot 5^2 \cdot 7^2} + \cdots.$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. For $n \geq 2$, $a_n = \frac{2^2 \cdot 4^2 \cdots (2n-2)^2}{3^2 \cdot 5^2 \cdots (2n-1)^2}$. Then

$$\frac{a_n}{a_{n+1}} = \frac{(2n+1)^2}{(2n)^2} = \frac{4n^2+4n+1}{4n^2} = 1 + \frac{1}{n} + \frac{1}{4n^2}.$$

Thus $\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + \frac{b_n}{n^2}$, where $\alpha = 1$ and $b_n = \frac{1}{4}$ is bounded. Hence, by Gauss's Test, the series diverges. □

Definition 3.6. *Let $f, \phi : \mathbb{N} \to \mathbb{R}$, where $\phi(n) > 0$ for all sufficiently large n. We write*

$$f(n) = O(\phi(n))$$

if there exist constants $k > 0$ and $N \in \mathbb{N}$ such that

$$|f(n)| \leq k\,\phi(n), \qquad \forall\, n \geq N.$$

In other words, for sufficiently large n, the function $f(n)$ does not grow faster than a constant multiple of $\phi(n)$. In particular, $O(1)$ denotes the class of all bounded functions.

Example 3.31. *Show that the polynomial $f(n) = 5n^2 + 3n + 1$ is of order $O(n^2)$.*

Proof. For $n \geq 1$, $3n \leq 3n^2$, $\quad 1 \leq n^2$. Hence,

$$f(n) = 5n^2 + 3n + 1 \leq 5n^2 + 3n^2 + n^2 = 9n^2.$$

Thus, $|f(n)| \leq 9n^2$, $\forall\, n \geq 1$. Therefore $f(n) = O(n^2)$. □

Theorem 3.32 (**Alternative Form of Gauss's Test**)**.** *Let $\sum_{n=1}^{\infty} a_n$ be a series of positive real numbers. If*

$$\frac{a_n}{a_{n+1}} = 1 + \frac{\alpha}{n} + O\left(\frac{1}{n^p}\right), p > 1,$$

then the series converges if $\alpha > 1$ and diverges if $\alpha \leq 1$.

Example 3.33. *Test the convergence of the series*

$$\left(\frac{1}{2}\right)^2 + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right)^2 + \cdots.$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. Then $a_n = \left(\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n}\right)^2$. Hence,

$$\frac{a_n}{a_{n+1}} = \left(\frac{2n+2}{2n+1}\right)^2 = \left(1 + \frac{1}{n}\right)^2 \left(1 + \frac{1}{2n}\right)^{-2}.$$

Using the expansion

$$\left(1 + \frac{1}{2n}\right)^{-2} = 1 - \frac{1}{n} + O\left(\frac{1}{n^2}\right),$$

we obtain $\frac{a_n}{a_{n+1}} = 1 + \frac{1}{n} + O\left(\frac{1}{n^2}\right)$. Therefore, $\alpha = 1$, and by the alternative form of Gauss's Test, the series diverges. □

Exercise 3.2. *Test the convergence of the following series.*

1. $$1 + \frac{1}{2}x + \frac{2^2}{3!}x^2 + \frac{3^2}{4!}x^3 + \cdots, \qquad x > 0.$$

2. $$\frac{3}{7} + \frac{3 \cdot 6}{7 \cdot 10}x + \frac{3 \cdot 6 \cdot 9}{7 \cdot 10 \cdot 13}x^2 + \cdots, \qquad x > 0.$$

3. $$\frac{1+x}{1!} + \frac{(1+2x)^2}{2!} + \frac{(1+3x)^3}{3!} + \cdots, \qquad x > 0.$$

4. $$1 + \frac{2^2}{3^2}x + \frac{2^2 \cdot 4^2}{3^2 \cdot 5^2}x^2 + \cdots, \qquad x > 0.$$

5. $$1 + \frac{1}{2!}x + \frac{(2!)^2}{4!}x^2 + \frac{(3!)^2}{6!}x^3 + \cdots, \qquad x > 0.$$

6. $$\frac{a}{b} + \frac{a(a+c)}{b(b+c)} + \frac{a(a+c)(a+2c)}{b(b+c)(b+2c)} + \cdots, \qquad a, b, c > 0.$$

3.3 Tests of convergence for Alternating Series

Definition 3.7. *A series of the form* $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ *where* (a_n) *is a sequence of positive terms is called an alternating series.*

Definition 3.8 (Absolutely Convergent Series)**.** *An infinite series* $\sum_{n=1}^{\infty} a_n$ *is said to be* ***absolutely convergent*** *if the series*

$$\sum_{n=1}^{\infty} |a_n|$$

is convergent.

Theorem 3.34. *An absolutely convergent series is convergent.*

Proof. Let $\sum_{n=1}^{\infty} u_n$ be a series of positive and negative real numbers and suppose that it is absolutely convergent. Then

$$\sum_{n=1}^{\infty} |u_n|$$

is a convergent series of positive terms.

Let $\varepsilon > 0$ be given. Since $\sum |u_n|$ converges, by Cauchy's criterion there exists a natural number m such that

$$|u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}| < \varepsilon$$

for all $n \geq m$ and every natural number p. Now,

$$\left|u_{n+1} + u_{n+2} + \cdots + u_{n+p}\right| \leq |u_{n+1}| + |u_{n+2}| + \cdots + |u_{n+p}| < \varepsilon,$$

for all $n \geq m$ and every natural number p. Therefore, by Cauchy's criterion, $\sum_{n=1}^{\infty} u_n$ is convergent.

□

Remark 3.35. *If $a_n \geq 0$ for every n, then $|a_n| = a_n$, and hence*

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} a_n.$$

Therefore, a series of non-negative terms is convergent if and only if it is absolutely convergent.

Example 3.36. *1. The series*

$$1 - \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} - \frac{1}{5^2} - \frac{1}{6^2} + \cdots$$

is convergent since it is absolutely convergent.

2. The series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

is convergent since it is absolutely convergent.

3. For a fixed value of x, the series $\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$ is absolutely convergent. Indeed,

$$\left|\frac{\sin(nx)}{n^2}\right| \leq \frac{1}{n^2},$$

and since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, the Comparison Test implies that $\sum_{n=1}^{\infty} \left|\frac{\sin(nx)}{n^2}\right|$ also converges. Hence, $\sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$ is absolutely convergent.

Theorem 3.37. *If the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent and $\{b_n\}$ is a bounded sequence, then the series*

$$\sum_{n=1}^{\infty} a_n b_n$$

is absolutely convergent.

Example 3.38. *Test the convergence of the series*

$$\frac{2}{1^3} - \frac{3}{2^3} + \frac{4}{3^3} - \frac{5}{4^3} + \cdots .$$

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series. Then

$$a_n = (-1)^{n+1}\frac{n+1}{n^3} = \frac{(-1)^{n+1}}{n^2}\left(1+\frac{1}{n}\right) = \alpha_n b_n,$$

where, $\alpha_n = \frac{(-1)^{n+1}}{n^2}$, $b_n = 1 + \frac{1}{n}$. Since $\sum_{n=1}^{\infty} \alpha_n$ is absolutely convergent and $\{b_n\}$ is a bounded sequence, it follows from the theorem that $\sum_{n=1}^{\infty} \alpha_n b_n$, that is, the given series, is absolutely convergent. □

Theorem 3.39 (Leibniz Test). *Let (a_n) be a decreasing sequence of non-negative real numbers such that $a_n \to 0$. Then the alternating series*

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converges. Moreover, if $S = \sum_{n=1}^{\infty} (-1)^{n-1} a_n$, then $a_1 - a_2 \leq S \leq a_1$.

Proof. Let $s_n = \sum_{k=1}^{n} (-1)^{k-1} a_k$. Observe that

$$s_{2n} = (a_1 - a_2) + \cdots + (a_{2n-1} - a_{2n}).$$

Since $a_{2n+1} \geq a_{2n+2}$,

$$s_{2n+2} - s_{2n} = a_{2n+1} - a_{2n+2} \geq 0.$$

Hence (s_{2n}) is increasing. Moreover,

$$s_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} \leq a_1.$$

Thus (s_{2n}) is bounded above, and therefore converges. Let $s_{2n} \longrightarrow S$. Now, $s_{2n+1} = s_{2n} + a_{2n+1}$. Since $a_{2n+1} \to 0$, we obtain $s_{2n+1} \longrightarrow S$. Hence, both the even and odd subsequences of (s_n) converge to the same limit. Therefore, $s_n \longrightarrow S$, which proves that the series converges. Finally,

$$a_1 - a_2 = s_2 \leq s_{2n} \leq S \leq s_{2n+1} \leq s_1 = a_1,$$

and letting $n \to \infty$ gives $a_1 - a_2 \leq S \leq a_1$. □

Example 3.40. *Consider the alternating harmonic series*

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Since the sequence $a_n = \frac{1}{n}$ is decreasing and satisfies $a_n \to 0$, the Leibniz Test implies that the series converges.

However, the series is not absolutely convergent, because

$$\sum_{n=1}^{\infty} \left|\frac{(-1)^{n-1}}{n}\right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

which is the harmonic series and is divergent (see Example 19). Thus, the alternating harmonic series is conditionally convergent.

Example 3.41. 1. *The alternating harmonic series*

$$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots$$

is convergent by the Leibniz Alternating Series Test.

2. *For $a>0$, the series*

$$\frac{1}{1+a^2}-\frac{1}{2+a^2}+\frac{1}{3+a^2}-\cdots$$

is convergent by the Leibniz Alternating Series Test.

3. *The series*

$$1-\frac{1}{2}+\frac{1\cdot 3}{2\cdot 4}-\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}+\cdots$$

is convergent by the Leibniz Alternating Series Test, since

$$\lim_{n\to\infty}\frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)}=0,$$

and

$$\frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots(2n)}>\frac{1\cdot 3\cdots(2n+1)}{2\cdot 4\cdots(2n+2)},$$

so the terms decrease monotonically to zero.

Theorem 3.42 (Abel's Test). *Let $\sum a_n$ be a convergent series, and let $\{b_n\}$ be a monotone bounded sequence. Then the series*

$$\sum_{n=1}^{\infty} a_n b_n$$

is convergent.

Example 3.43. 1. *The series*

$$\sum_{n=2}^{\infty}\frac{(-1)^{n+1}}{n\log n}$$

is convergent by Abel's Test, since $\sum\limits_{n=2}^{\infty}\frac{(-1)^{n+1}}{n}$ is convergent and the sequence $\left\{\frac{1}{\log n}\right\}_{n=2}^{\infty}$ is monotone decreasing and bounded below.

2. *The series*

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n+1}\left(1+\frac{1}{n}\right)^{-n}$$

is convergent by Abel's Test, since $\sum\limits_{n=1}^{\infty}\frac{(-1)^{n+1}}{n+1}$ is convergent and the sequence $\left\{\left(1+\frac{1}{n}\right)^{-n}\right\}_{n=1}^{\infty}$ is monotone decreasing and bounded below.

Definition 3.9 (Conditionally Convergent Series). *A series is said to be* ***conditionally convergent*** *if it is convergent but not absolutely convergent.*

Example 3.44. *Show that the series*

$$\frac{1}{(1+a)^p}-\frac{1}{(2+a)^p}+\frac{1}{(3+a)^p}-\cdots, \qquad a>0$$

is

1. *absolutely convergent if* $p > 1$,
2. *conditionally convergent if* $0 < p \leq 1$.

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the given series and let $b_n = |a_n|$. Then $\sum_{n=1}^{\infty} b_n$ is a series of positive real numbers and $b_n = \frac{1}{(n+a)^p}$. Let

$$c_n = \frac{1}{n^p}.$$

Then

$$\lim_{n\to\infty} \frac{b_n}{c_n} = \lim_{n\to\infty} \frac{n^p}{(n+a)^p} = 1.$$

By the Limit Comparison Test, $\sum_{n=1}^{\infty} b_n$ converges if $p > 1$ and diverges if $0 < p \leq 1$.

Case 1: $p > 1$.
In this case, $\sum_{n=1}^{\infty} b_n$ converges. Hence $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.

Case 2: $0 < p \leq 1$.
In this case, $b_n = \frac{1}{(n+a)^p}$ is a monotone decreasing sequence of positive real numbers and $\lim_{n\to\infty} b_n = 0$. Since the given series is alternating, by Leibniz's Test, $\sum_{n=1}^{\infty} a_n$ converges. But $\sum_{n=1}^{\infty} b_n$ diverges. Therefore, $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. □

Exercise 3.3. *Test the convergence of each of the following series.*

(i)
$$1 - \frac{1}{2!} + \frac{1}{4!} - \frac{1}{6!} + \cdots.$$

(ii)
$$\frac{1}{2^2 \log 2} - \frac{1}{3^2 \log 3} + \frac{1}{4^2 \log 4} - \cdots.$$

(iii)
$$\frac{x}{1+x} - \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} - \cdots, \qquad 0 < x < 1.$$

(iv)
$$1 - \frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \cdots.$$

(v)
$$\frac{1}{(\log 2)^2} - \frac{1}{(\log 3)^2} + \frac{1}{(\log 4)^2} - \cdots.$$

4 Problem Sheet

Problems on Sequences

Exercise 4.1. 1. *Discuss the convergence of the following sequences:*

(i) $\left\{\frac{(-1)^n n}{n^2+1}\right\}$

(ii) $\left\{\frac{3n^2+\sin n-4}{2n^2+3}\right\}$

(iii) $\left\{\frac{n+1}{\sqrt{n}}\right\}$

(iv) $\left\{\left(1+\frac{1}{n^2}\right)^{n^2}\right\}$

(v) $\left\{\left(\left(1+\frac{1}{n^2}\right)\left(1+\frac{2}{n^2}\right)\left(1+\frac{3}{n^2}\right)\right)^{n^2}\right\}$

2. *Determine whether the following sequences are increasing, decreasing, or neither:*

(i) $\left\{\frac{3^n}{3^{n+1}}\right\}$ (ii) $\left\{\frac{5n-2}{4n+1}\right\}$ (iii) $\left\{\cos\frac{n\pi}{3}\right\}$

3. *Give an example of divergent sequences $\{x_n\}$ and $\{y_n\}$ such that $\{x_n y_n\}$, $\{x_n + y_n\}$ is convergent.*

4. *If $x_n > 0$ for all $n \in \mathbb{N}$ and $\lim \sqrt[n]{x_n} = l > 0$, show that $\lim \sqrt[n]{(n+1)x_{n+1}} = l$.*

5. *Examine whether the sequence $\{x_n\}$ where $x_n = 1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{n!}$ is convergent or not. If $\{x_n\}$ is convergent, find its limit.*

6. *Show that the sequence $\{1, \frac{1}{2}, 1, \frac{2}{3}, 1, \frac{3}{4}, \dots\}$ converges to 1.*

7. *Two sequences $\{x_n\}$ and $\{y_n\}$ are defined by $x_{n+1} = \frac{1}{2}(x_n + y_n)$, $y_{n+1} = \sqrt{(x_n + y_n)}$ for $n \geq 1$ and $x_1 > 0$, $y_1 > 0$. Prove that both sequences converge to a common limit.*

8. *Give an example of a sequence of rational numbers that converges to an irrational number. Also give an example of a sequence of irrational numbers that converges to a rational number.*

9. *If the sequence $\{x_n\}$ is defined as $x_1 = 1$ and $x_{n+1} = \frac{1}{x_n+2}$ for all $n \in \mathbb{N}$, prove that $\{x_n\}$ is convergent and find its limit.*

10. *Find the limit of the sequence* $\left\{\sqrt[n]{\frac{2n!}{n!n!}}\right\}$.

11. *Find the limit of the sequence* $\left\{\frac{1}{n}\sqrt[n]{(2n+1)(2n+2)\cdots(2n+n)}\right\}$.

12. *Which of the following statements are correct?*

(a) $n\log\left(1+\dfrac{1}{n+1}\right) \to 1$ *as* $n \to \infty$.

(b) $(n+1)\log\left(1+\dfrac{1}{n}\right) \to 1$ *as* $n \to \infty$.

(c) $n^2\log\left(1+\dfrac{1}{n}\right) \to 1$ *as* $n \to \infty$.

(d) $n \log\left(1+\frac{1}{n^2}\right) \to 1 \quad$ *as* $n \to \infty$.

Prove that the sequence $\{u_n\}$ *defined by the following recurrence relations converges, and find its limit.*

(a) *If* $0 < u_1 < u_2$ *and* $u_{n+2} = \frac{2u_{n+1}+u_n}{3}$, $n \geq 1$, *then show that* u_n *converges to* $\frac{u_1+3u_2}{4}$.

(b) *If* $0 < u_1 < u_2$ *and* $\frac{2}{u_{n+2}} = \frac{1}{u_{n+1}} + \frac{1}{u_n}$, ≥ 1, *then show that* u_n *converges*

$$\frac{3}{\dfrac{1}{u_1}+\dfrac{2}{u_2}}.$$

13. *If* $s_1 > 0$ *and* $s_{n+1} = \frac{1}{2}\left(s_n + \frac{4}{s_n}\right)$, $n \geq 1$, *prove that the sequence* $\{s_n\}$ *is monotone decreasing, bounded below, and* $\lim\limits_{n\to\infty} s_n = 2$.

Problems on Series

Exercise 4.2. 1. *If* $\sum\limits_{n=1}^{\infty} x_n$ *is a convergent series of positive real numbers, prove that*

(a) $\sum\limits_{n=1}^{\infty} x_n^2$ *is convergent.*

(b) $\sum\limits_{n=1}^{\infty} \frac{x_n}{n}$ *is convergent.*

(c) $\sum\limits_{n=1}^{\infty} x_{2n}$ *is convergent.*

(d) $\sum\limits_{n=1}^{\infty} \frac{x_n}{1+x_n}$ *is convergent.*

2. *Let* $\{a_n\}_{n\geq 1}$ *be a sequence of real numbers satisfying* $a_1 \geq 1$ *and* $a_n \geq a_{n+1}$ *for all* $n \geq 1$. *Then which of the following is necessarily true?*

(a) *The series* $\sum\limits_{n=1}^{\infty} \frac{1}{a_n}$ *diverges.*

(b) *The sequence* $\{a_n\}_{n\geq 1}$ *is bounded.*

(c) *The series* $\sum\limits_{n=1}^{\infty} \frac{1}{a_n^2}$ *converges.*

(d) *The series* $\sum\limits_{n=1}^{\infty} \frac{1}{a_n}$ *converges.*

3. *Discuss the convergence/divergence of the following series.*

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2+1}{(n+3)(n+4)}$. (b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{10n^2+n-13}$. (c) $\displaystyle\sum_{n=1}^{\infty} \frac{\log n}{2n^3-1}$.

(d) $\displaystyle\sum_{n=1}^{\infty} \sin^2 \frac{1}{n}$. (e) $\displaystyle\sum_{n=1}^{\infty} n \tan \frac{1}{n^2}$. (f) $\displaystyle\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$.

(g) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{n^2+1}$ (h) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n 3^n}{n!}$ (i) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(n!)^2}{(2n)!} 5^n$

(j) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \log(n+1)}{n+1}$. (k) $\displaystyle\sum_{n=1}^{\infty} \left[\sqrt[3]{(n^3+1)} - n\right]$.

4. *For* $a > 0$, *the series* $\displaystyle\sum_{n=1}^{\infty} a^{\ln n}$ *is convergent if and only if*

(a) $0 < a < e$
(b) $0 < a \leq e$
(c) $0 < a < \dfrac{1}{e}$
(d) $0 < a \leq \dfrac{1}{e}$

5. *Using the fact that* $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \log 2$, *the series* $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)}$ *equals*

(a) $1 - 2\log 2$
(b) $1 + \log 2$
(c) $(\log 2)^2$
(d) $-(\log 2)^2$

6. *The sum of the series* $\displaystyle\sum_{k=1}^{\infty} \frac{1}{(2k-1)^2}$ *is*

(a) $\dfrac{\pi^2}{8}$
(b) $\dfrac{\pi^2}{6}$
(c) $\dfrac{\pi}{2}$
(d) 1

7. *The series* $\displaystyle\sum_{k=0}^{\infty} \frac{k^2+3k+1}{(k+2)!}$ *converges to*

(a) 1
(b) $\dfrac{1}{2}$

(c) 2

(d) 3

8. *We know that* $xe^x = \sum_{n=1}^{\infty} \frac{x^n}{(n-1)!}$. *The series* $\sum_{n=1}^{\infty} \frac{2^n n^2}{n!}$ *converges to*

   (a) e^2

   (b) $2e^2$

   (c) $4e^2$

   (d) $6e^2$

9. *The infinite series* $\sum_{n=1}^{\infty} \frac{a^n \log n}{n^2}$ *converges if and only if*

   (a) $a \in [-1, 1)$

   (b) $a \in (-1, 1]$

   (c) $a \in [-1, 1]$

   (d) $a \in (-\infty, \infty)$

10. *Discuss the convergence of* $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n\sqrt{n}}$ *and* $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{n^2+1}$.

11. *Prove that* $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n(\log n)^{1/3}}$ *is conditionally convergent.*

12. *Prove that the series*

$$\left(\frac{1}{2}\right)^2 - \left(\frac{1\cdot 4}{2\cdot 5}\right)^2 + \left(\frac{1\cdot 4\cdot 7}{2\cdot 5\cdot 8}\right)^2 - \cdots$$

is conditionally convergent.

4.1 Limits

In many situations, the value of a function at a point is less important than its behavior near that point. For example, consider $f(x) = \frac{x^2-1}{x-1}$, $x \neq 1$. Although $f(1)$ is not defined, we have $f(x) = x + 1$, $x \neq 1$, so the values of $f(x)$ become arbitrarily close to 2 as x approaches 1. Thus, the function exhibits a well-defined behavior near $x = 1$ despite being undefined there.

More generally, when studying continuous and discontinuous functions, we are interested in understanding the behavior of a function as the input approaches a given point from the left or the right. This motivates the concept of the *limit of a function*, which provides a precise mathematical description of the value that a function approaches as its argument approaches a specified point.

Definition 4.1. *Let $a \in J \subseteq \mathbb{R}$ be an interval and $f : J \setminus \{a\} \to \mathbb{R}$ be a function. We say that*

$$\lim_{x \to a} f(x) = \ell$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

In this case, we say that ℓ is the limit of f at a.

Example 4.1. *Show that*

1. $\lim\limits_{x \to 1} \frac{x^2-1}{x-1} = 2.$
2. $\lim\limits_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0.$

Solution.

1. *Let $\epsilon > 0$ be given. Choose $\delta = \epsilon$. If $0 < |x - 1| < \delta$, then*

$$\left| \frac{x^2 - 1}{x - 1} - 2 \right| = |x + 1 - 2| = |x - 1| < \delta = \epsilon.$$

   Hence, $\lim\limits_{x \to 1} \frac{x^2-1}{x-1} = 2.$

2. *Let $\epsilon > 0$ be given and choose $\delta = \epsilon$. If $0 < |x| < \delta$, then*

$$\left| x \sin\left(\frac{1}{x}\right) \right| = |x| \left| \sin\left(\frac{1}{x}\right) \right| \leq |x| < \delta = \epsilon.$$

   Hence, $\lim\limits_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0.$

Theorem 4.2. *The limit of a function, if it exists, is unique.*

Proof. Suppose $\lim\limits_{x \to a} f(x) = \ell_1$ and $\lim\limits_{x \to a} f(x) = \ell_2$. Assume, to the contrary, that $\ell_1 \neq \ell_2$. Choose $\epsilon = \frac{|\ell_1 - \ell_2|}{3} > 0$. Then there exist $\delta_1, \delta_2 > 0$ such that

$$0 < |x - a| < \delta_j \Longrightarrow |f(x) - \ell_j| < \epsilon, \qquad j = 1, 2.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. If $0 < |x - a| < \delta$, then

$$|f(x) - \ell_1| < \epsilon \quad \text{and} \quad |f(x) - \ell_2| < \epsilon.$$

Hence,

$$|\ell_1 - \ell_2| \leq |\ell_1 - f(x)| + |f(x) - \ell_2| < 2\epsilon = \frac{2}{3}|\ell_1 - \ell_2|,$$

which is impossible. Therefore, $\ell_1 = \ell_2$. $\square$

Theorem 4.3 (**Algebra of Finite Limits**). *Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$, and let $f, g : J \setminus \{a\} \to \mathbb{R}$ be functions. Assume that*

$$\lim_{x \to a} f(x) = \ell_1 \quad \text{and} \quad \lim_{x \to a} g(x) = \ell_2,$$

where $\ell_1, \ell_2 \in \mathbb{R}$. Let $\alpha \in \mathbb{R}$. Then:

1. $\lim_{x \to a} (f(x) + g(x)) = \ell_1 + \ell_2$.
2. $\lim_{x \to a} (\alpha f(x)) = \alpha \ell_1$.
3. $\lim_{x \to a} (f(x)g(x)) = \ell_1 \ell_2$.
4. *If $\ell_1 \neq 0$, then there exists $\delta > 0$ such that $f(x) \neq 0$ whenever $0 < |x - a| < \delta$. Moreover,*

$$\lim_{x \to a} \frac{1}{f(x)} = \frac{1}{\ell_1}.$$

Definition 4.2 (**Right-Hand Limit**). *Let $J \subseteq \mathbb{R}$ be an interval and $a \in J$ be not the right endpoint of J. Suppose $f : J \setminus \{a\} \to \mathbb{R}$ be a function. We say that*

$$\lim_{x \to a^+} f(x) = \ell$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < x - a < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

*In this case, ℓ is called the **right-hand limit** of f at a.*

Definition 4.3 (**Left-Hand Limit**). *Let $J \subseteq \mathbb{R}$ be an interval and $a \in J$ be not the left endpoint of J. Suppose $f : J \setminus \{a\} \to \mathbb{R}$ be a function. We say that*

$$\lim_{x \to a^-} f(x) = \ell$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < a - x < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

*In this case, ℓ is called the **left-hand limit** of f at a.*

Example 4.4. *Let*

$$f(x) = \begin{cases} 0, & \text{if } x > 0, \\ 1, & \text{if } x < 0. \end{cases}$$

Then

$$\lim_{x \to 0^+} f(x) = 0, \qquad \text{and} \qquad \lim_{x \to 0^-} f(x) = 1.$$

Theorem 4.5. *Let $J \subseteq \mathbb{R}$ be an interval and $a \in J$ be neither the left endpoint nor the right endpoint of J. Suppose $f : J \setminus \{a\} \to \mathbb{R}$ be a function. Then*

$$\lim_{x \to a} f(x) = \ell$$

if and only if

$$\lim_{x \to a^-} f(x) = \ell \quad \text{and} \quad \lim_{x \to a^+} f(x) = \ell.$$

Theorem 4.6 (Sequential Characterization of Limits). *Let $a \in J \subseteq \mathbb{R}$ be an interval and $f : J \setminus \{a\} \to \mathbb{R}$ be a function. Then $\lim\limits_{x \to a} f(x) = \ell$ if and only if for every sequence (x_n) in $J \setminus \{a\}$ satisfying $x_n \to a$, we have $f(x_n) \to \ell$, as $n \to \infty$.*

Example 4.7. *Show that*

$$\lim_{x \to 2} (x^2 + 1) = 5.$$

Solution. *Let (x_n) be any sequence such that $x_n \to 2$. Then, by the algebra of convergent sequences,*

$$x_n^2 + 1 \to 2^2 + 1 = 5.$$

Hence, by the Sequential Characterization of Limits,

$$\lim_{x \to 2} (x^2 + 1) = 5.$$

Exercise 4.3. *Let*

$$f(x) = \begin{cases} 0, & \text{if } x \in \mathbb{Q}, \\ 1, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Show that

$$\lim_{x \to a} f(x)$$

exists only for $a = 0$. Hint. Use the density of the rational and irrational numbers in $\mathbb{R}$. Alternatively, use the Sequential Characterization of Limits by considering one sequence of rational numbers and one sequence of irrational numbers, both converging to a.

4.2 Infinite Limits

Definition 4.4 (Infinite Limits). *Let $a \in J \subseteq \mathbb{R}$ be an interval and $f : J \setminus \{a\} \to \mathbb{R}$ be a function.*

1. *We say that*

   $$\lim_{x \to a} f(x) = +\infty$$

   if for every $M > 0$, there exists $\delta > 0$ such that

   $$x \in J, \quad 0 < |x - a| < \delta \Rightarrow f(x) > M.$$

2. *We say that*

   $$\lim_{x \to a} f(x) = -\infty$$

if for every $M < 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow f(x) < M.$$

Example 4.8. *Show that* $\lim\limits_{x \to 2} \frac{1}{(x-2)^2} = +\infty$.

Solution. *Let $M > 0$ be given. Choose $\delta = \frac{1}{\sqrt{M}}$. If $0 < |x-2| < \delta$, then*

$$\frac{1}{(x-2)^2} > M.$$

Hence, by the definition of an infinite limit,

$$\lim_{x \to 2} \frac{1}{(x-2)^2} = +\infty.$$

Exercise 4.4. *Solve the following exercises.*

1. *Use the sequential criterion for limits to show that the following limits do not exist.*

   (i) $\lim\limits_{x \to 0} \cos \frac{\pi}{x}$. *(ii)* $\lim\limits_{x \to 0} \frac{1}{x} \sin \frac{1}{x}$.

2. *Let*

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ -2 - x, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

   Show that

   (i) $\lim\limits_{x \to -1} f(x) = 1$. *(ii)* $\lim\limits_{x \to c} f(x)$

   does not exist whenever $c \neq -1$.

3. *Show that the following limits do not exist.*

   (i) $\lim\limits_{x \to 0} \frac{\sin |x|}{\sin x}$. *(ii)* $\lim\limits_{x \to 0} \frac{2x + |x|}{x^2 - |x|}$.

4. *Evaluate the following limits.*

   (i) $\lim\limits_{x \to 0^+} \sqrt{x - [x]}$, $\lim\limits_{x \to 0^-} \sqrt{x - [x]}$. *(iii)* $\lim\limits_{x \to 0^+} \big([x] + [1-x]\big)$, $\lim\limits_{x \to 0^-} \big([x] + [1-x]\big)$.

   (ii) $\lim\limits_{x \to 0^+} \frac{1}{e^x + \frac{1}{x}}$, $\lim\limits_{x \to 0^-} \frac{1}{e^x + \frac{1}{x}}$. *(iv)* $\lim\limits_{x \to 0^+} x \left[\frac{1}{x}\right]$, $\lim\limits_{x \to 0^-} x \left[\frac{1}{x}\right]$.

5. *Evaluate the following limits.*

   (i) $\lim\limits_{x \to \infty} \frac{\sin x}{e^x \cos x}$. *(ii)* $\lim\limits_{x \to \infty} \left(x - \sqrt{x^2 - 1}\right)$. *(iii)* $\lim\limits_{x \to \infty} \left(\sqrt[3]{x+1} - \sqrt[3]{x}\right)$.

5 Continuity

The concept of a limit describes the behavior of a function as the input approaches a point. A natural question is whether the value of the function at that point agrees with the value suggested by its nearby behavior. If this happens at every point in its domain, the function is said to be *continuous*.

Intuitively, a continuous function has no sudden jumps, breaks, or holes in its graph. Small changes in the input produce only small changes in the output. This idea is fundamental in mathematics and its applications, as many physical phenomena, such as the motion of an object, the flow of heat, and population growth, are naturally modeled by continuous functions. The precise definition of continuity is therefore expressed in terms of limits.

Definition 5.1. *Let $a \in J \subseteq \mathbb{R}$ and $f : J \to \mathbb{R}$ be a function. We say that f is* ***continuous at*** *a if for every $\epsilon > 0$, there exists $\delta > 0$ such that*

$$x \in J \text{ and } |x - a| < \delta \quad \Longrightarrow \quad |f(x) - f(a)| < \epsilon.$$

We say that f is ***continuous on*** *J if it is continuous at every point of J.*

Example 5.1. *Show that the following functions are continuous.*

1. $f : \mathbb{R} \to \mathbb{R},\ f(x) = x$, *for all* $x \in \mathbb{R}$.
2. $f : [-R, R] \to \mathbb{R},\ f(x) = x^2$, *for all* $x \in [-R, R]$.

Solution.

1. *Let $a \in \mathbb{R}$ and let $\epsilon > 0$ be given. We have*
$$|f(x) - f(a)| = |x - a|.$$
Thus, it is natural to choose
$$\delta = \epsilon.$$
Then, whenever $|x - a| < \delta$, we obtain
$$|f(x) - f(a)| = |x - a| < \delta = \epsilon.$$
Hence, f is continuous at a. Since a was arbitrary, f is continuous on $\mathbb{R}$.

2. *Let $a \in [-R, R]$ and let $\epsilon > 0$ be given. Observe that*
$$|f(x) - f(a)| = |x^2 - a^2| = |x + a||x - a| \le (|x| + |a|)|x - a| \le 2R|x - a|.$$
Choose
$$\delta = \frac{\epsilon}{2R}.$$
Then, whenever $|x - a| < \delta$, we have
$$|f(x) - f(a)| \le 2R|x - a| < 2R\delta = \epsilon.$$
Hence, f is continuous on $[-R, R]$.

Exercise 5.1. *Show that the function $f : \mathbb{R} \to \mathbb{R}$, defined by $f(x) = x^2$, is continuous on $\mathbb{R}$ using the ϵ-δ definition. Hint. Observe that $|x^2 - a^2| = |x-a||x+a|$. To estimate $|x+a|$, first assume $\delta \le 1$. Then, whenever $|x - a| < \delta$, we have*

$$|x| \le |x - a| + |a| < 1 + |a|,$$

and hence

$$|x + a| \le |x| + |a| < 1 + 2|a|.$$

Now, choose $\delta = \min\left\{1, \frac{\epsilon}{1+2|a|}\right\}$.

5.1 Sequential Criterion for Continuity

Definition 5.2. *Let $a \in J \subseteq \mathbb{R}$ and $f : J \to \mathbb{R}$ be a function. We say that f is **continuous at** a if for every sequence (x_n) in J with $x_n \to a$, we have $f(x_n) \to f(a)$. We say that f is **continuous on** J if it is continuous at every point of J.*

Example 5.2. *1. Show that every constant function is continuous.*

2. Show that the identity function $f : J \to \mathbb{R}$, defined by $f(x) = x$ for all $x \in J$, is continuous.

3. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then f is not continuous at 0.

Solution.

1. *Let $J \subseteq \mathbb{R}$ and $f : J \to \mathbb{R}$ be the constant function defined by $f(x) = c$ for all $x \in J$.*

   Fix any $a \in J$, and (x_n) be a sequence in J such that $x_n \to a$. Since $f(x_n) = c$ for every n, the sequence $(f(x_n))$ is constant. Hence,

   $$f(x_n) \to c = f(a).$$

   Therefore, f is continuous at a. As a was arbitrary, we conclude that f is continuous on J.

2. *Fix any $a \in J$, and let (x_n) be a sequence in J such that $x_n \to a$. Since $f(x_n) = x_n$ for every n, we have*

   $$f(x_n) = x_n \longrightarrow a = f(a).$$

   Hence, f is continuous at a. Since a was arbitrary, f is continuous on J.

   More generally, for every $n \in \mathbb{N}$, the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^n$ is continuous.

3. *Consider the sequence $x_n = \frac{1}{n}$. Then $x_n \in \mathbb{Q}$ and $x_n \to 0$. Hence*

   $$f(x_n) = 1 \longrightarrow 1 = f(0).$$

   Now let

   $$y_n = \frac{\sqrt{2}}{n}.$$

Since $\sqrt{2}$ is irrational, y_n is irrational for every n. Also, $y_n \to 0$. Therefore,

$$f(y_n) = 0 \longrightarrow 0 \neq 1 = f(0).$$

Thus, by the sequential criterion for continuity, f is not continuous at 0. Using the density of the rational and irrational numbers in $\mathbb{R}$, together with the Sandwich Theorem, one can similarly show that f is not continuous at any point of $\mathbb{R}$.

Example 5.3. *Consider the function*

$$f(x) = \begin{cases} 1, & \text{if } x > 0, \\ 0, & \text{if } x \leq 0. \end{cases}$$

We claim that f is continuous at every $a \neq 0$ but is not continuous at 0. Indeed, let $a \neq 0$ and let (x_n) be a sequence converging to a. Since a is either positive or negative, there exists $N \in \mathbb{N}$ such that x_n has the same sign as a for all $n \geq N$. Consequently, $f(x_n) = f(a)$ for all sufficiently large n, and hence

$$f(x_n) \longrightarrow f(a).$$

Thus f is continuous at every $a \neq 0$. Now let $a = 0$. Consider the sequences

$$x_n = \frac{1}{n}, \qquad y_n = -\frac{1}{n}.$$

Both converge to 0, but

$$f(x_n) = 1 \longrightarrow 1, \qquad f(y_n) = 0 \longrightarrow 0.$$

Since the limits are different, f is not continuous at 0.

Example 5.4. *Let $f : \mathbb{R} \to \mathbb{R}$ be defined by*

$$f(x) = \begin{cases} c, & \text{if } x < 0, \\ \alpha, & \text{if } x = 0, \\ c, & \text{if } x > 0. \end{cases}$$

Determine the value of α for which f is continuous at 0.

Proof. Suppose that f is continuous at 0. Consider the sequence

$$x_n = -\frac{1}{n}.$$

Then $x_n \to 0$ and

$$f(x_n) = \alpha \longrightarrow f(0) = c.$$

Hence,

$$\alpha = c.$$

Conversely, assume that $\alpha = c$. Let (x_n) be any sequence with $x_n \to 0$. If $x_n < 0$, then

$$f(x_n) = c = f(0).$$

If $x_n > 0$, then

$$f(x_n) = c = f(0).$$

In either case, $f(x_n) \to c = f(0)$. Hence, f is continuous at 0. Thus, f is continuous at 0 if and only if $\alpha = c$. ☐

Theorem 5.5. *The sequential definition of continuity and the ϵ-δ definition are equivalent.*

Proof. Left as an exercise. $\square$

Theorem 5.6 (Algebra of Continuous Functions). *Let $J \subseteq \mathbb{R}$ and $f, g : J \to \mathbb{R}$ be continuous at $a \in J$. Suppose $\alpha \in \mathbb{R}$. Then:*

1. *$f + g$ is continuous at a.*
2. *αf is continuous at a.*
3. *fg is continuous at a.*
4. *If $g(a) \neq 0$, then there exists $\delta > 0$ such that $g(x) \neq 0$ for all $x \in (a - \delta, a + \delta) \cap J$. Consequently, $\frac{1}{g}$ is continuous at a, and hence so is $\frac{f}{g}$.*
5. *$|f|$ is continuous at a.*
6. *The functions $\max\{f, g\}$ and $\min\{f, g\}$ are continuous at a.*
7. *If $K \subseteq \mathbb{R}$, $h : K \to \mathbb{R}$ is continuous at $f(a)$, and $f(J) \subseteq K$, then the composition $h \circ f$ is continuous at a.*

Remark 5.7. *If $|f|$ is continuous on J, then f need not be continuous on J. For example, let $J = \mathbb{R}$ and define*

$$f : \mathbb{R} \to \mathbb{R}, \qquad f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ -1, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Then $|f| : \mathbb{R} \to \mathbb{R}$ is given by $|f|(x) = 1,\ x \in \mathbb{R}$. Hence $|f|$ is continuous on $\mathbb{R}$, whereas f is not continuous on $\mathbb{R}$.

6 Discontinuity

We have seen that a function defined on a subset J of $\mathbb{R}$ may be continuous at every point of J, continuous only at some points, or discontinuous at one or more points of J. The behavior of a function near a point determines the nature of its discontinuity.

Different discontinuities arise in different ways. To understand and classify them, we now study the various types of discontinuities of a function.

6.1 Types of Discontinuities

1. $\lim_{x \to a^-} f(x)$ does not exist.
2. $\lim_{x \to a^+} f(x)$ does not exist.
3. $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$: **Jump discontinuity**
4. $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) \neq f(a)$: **Removable discontinuity**

Example 6.1. *1. Consider the function*

$$f(x) = \sin\left(\frac{1}{x}\right), \qquad x \neq 0.$$

Then $\lim_{x \to 0^-} f(x)$ does not exist.

2. *Consider the function*

$$f(x) = \frac{1}{x}, \qquad x \neq 0.$$

Then $\lim_{x \to 0^-} f(x)$ *does not exist (it is* $-\infty$*).*

3. *Consider the function*

$$f(x) = \begin{cases} 1, & \text{if } x > 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then

$$\lim_{x \to 0^-} f(x) = 0 \quad \text{and} \quad \lim_{x \to 0^+} f(x) = 1.$$

Since the left-hand and right-hand limits are different, f *is discontinuous at* 0*.*

4. *Consider the function*

$$f(x) = \begin{cases} 1, & \text{if } x \neq 0, \\ 2026, & \text{if } x = 0. \end{cases}$$

Then

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) = 1,$$

but $f(0) = 2026$*. Hence* f *is discontinuous at* 0*.*

5. *Consider the function*

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Since every interval contains both rational and irrational numbers, neither

$$\lim_{x \to 0^-} f(x) \quad \text{nor} \quad \lim_{x \to 0^+} f(x)$$

exists. Hence f *is discontinuous at* 0*. Indeed, the function is discontinuous at every point of* $\mathbb{R}$*.*

6. *Define*

$$f(x) = \begin{cases} x, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then f *is continuous only at* $x = 0$ *and is discontinuous at every other point of* $\mathbb{R}$*.*

6.2 Properties of Continuous Functions

Suppose $f : [a, b] \to \mathbb{R}$ is a continuous function such that $f(a)$ and $f(b)$ have opposite signs. Draw the graphs of a few such functions, considering both the cases $f(a) > 0 > f(b)$ and $f(a) < 0 < f(b)$.

What do you observe? Does the graph always intersect the x-axis? If so, can you explain why this must happen? In other words, must there exist a point $c \in (a, b)$ such that $f(c) = 0$?

The answer is affirmative, and it is one of the most important consequences of continuity. This result is known as the Intermediate Value Theorem.

Theorem 6.2 (Cantor's Nested Interval Theorem)**.** *Let* $\{[a_n, b_n]\}$ *be a sequence of closed intervals such that*

1. $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ *for all* $n \in \mathbb{N}$*,*

2. $\lim\limits_{n\to\infty}(b_n - a_n) = 0$.

Then $\bigcap\limits_{n=1}^{\infty}[a_n, b_n]$ *contains exactly one point.*

Theorem 6.3 (Bolzano)**.** *Let* $[a,b]$ *be a closed and bounded interval and* $f : [a,b] \to \mathbb{R}$ *be continuous on* $[a,b]$*. If* $f(a)$ *and* $f(b)$ *are of opposite signs then there exists at least a point* c *in the open interval* (a,b) *such that* $f(c) = 0$*.*

Proof. Let $I_1 = [a,b] = [a_1, b_1]$, say. Without loss of generality, let us assume that $f(a_1) < 0$ and $f(b_1) > 0$.

Let $c_1 = \frac{a_1+b_1}{2}$. Then $f(c_1)$ is either 0 or $\neq 0$. If $f(c_1) = 0$ the theorem is proved. If $f(c_1) \neq 0$ then either $f(c_1) < 0$, or $f(c_1) > 0$.

If $f(c_1) < 0$ we consider the closed interval $[c_1, b_1]$ and call it $I_2 = [a_2, b_2]$. If $f(c_1) > 0$ we consider the closed interval $[a_1, c_1]$ and call it $I_2 = [a_2, b_2]$.

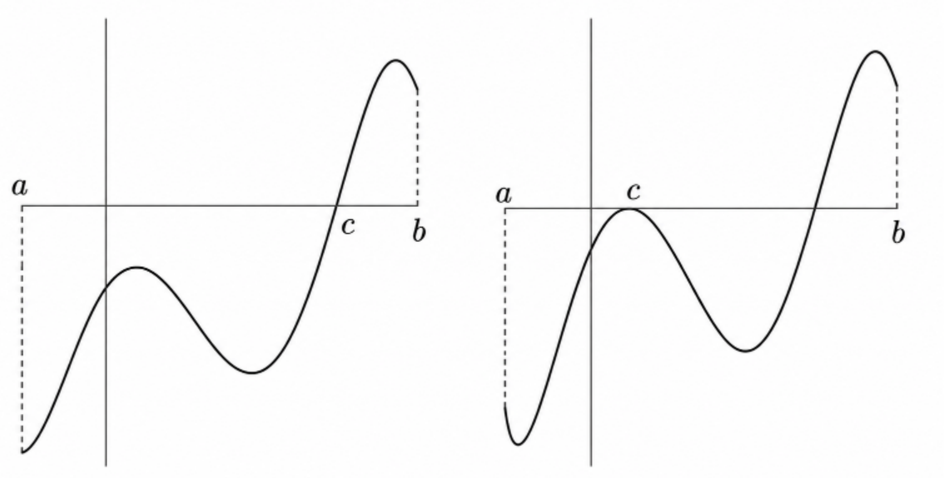
Thus if $f(c_1) \neq 0$, the closed interval $I_2 = [a_2, b_2]$ is such that (i) f is continuous on I_2 and $f(a_2) < 0$, $f(b_2) > 0$; (ii) $I_2 \subset I_1$; (iii) $|I_2| = \frac{1}{2}(b-a)$.

Let $c_2 = \frac{a_2+b_2}{2}$. Then $f(c_2)$ is either 0 or $\neq 0$.

If $f(c_2) = 0$ the theorem is proved. If $f(c_2) \neq 0$, then either $f(c_2) < 0$, or $f(c_2) > 0$.

If $f(c_2) < 0$ we consider the closed interval $[c_2, b_2]$ and call it $I_3 = [a_3, b_3]$. If $f(c_2) > 0$ we consider the closed interval $[a_2, c_2]$ and call it $I_3 = [a_3, b_3]$.

Thus if $f(c_2) \neq 0$ the closed interval $I_3 = [a_3, b_3]$ is such that (i) f is continuous on I_3 and $f(a_3) < 0$, $f(b_3) > 0$; (ii) $I_3 \subset I_2 \subset I_1$; (iii) $|I_3| = \frac{b-a}{2}$.



Let $c_3 = \frac{a_3+b_3}{2}$. Then $f(c_3)$ is either 0 or $\neq 0$.

Continuing in this manner, either we obtain a point c_k in (a,b) such that $f(c_k) = 0$, in which case the theorem is proved, or we obtain a sequence of closed and bounded intervals $\{I_n\}$ such that (i) f is continuous on I_n and $f(a_n) < 0$, $f(b_n) > 0$ for all $n \in \mathbb{N}$; (ii) $I_{n+1} \subset I_n$ for all $n \in \mathbb{N}$; (iii) $|I_n| = \frac{b-a}{2^n}$ and therefore $\lim |I_n| = 0$.

Thus $\{I_n\}$ is a sequence of nested closed and bounded intervals with $\lim |I_n| = 0$.

By nested intervals theorem there exists one and only one point α such that (i) $\alpha \in [a_n, b_n]$ for all $n \in \mathbb{N}$, and (ii) $\lim a_n = \alpha = \lim b_n$.

From (i) it follows that $\alpha \in [a,b]$. Since f is continuous on $[a,b]$, f is continuous at α.

From (ii) it follows that $\{a_n\}$ is a sequence of points in $[a, b]$ converging to α. Since f is continuous at α, $\lim f(a_n) = f(\alpha)$.

But $f(a_n) < 0$ for all $n \in \mathbb{N}$ and this implies $\lim f(a_n) \le 0$.

That is,

$$f(\alpha) \le 0 \tag{A}$$

Also from (ii) it follows that $\{b_n\}$ is a sequence of points in $[a, b]$ converging to α. Since f is continuous at α, $\lim f(b_n) = f(\alpha)$.

But $f(b_n) > 0$ for all $n \in \mathbb{N}$ and this implies $\lim f(b_n) \ge 0$.

That is,

$$f(\alpha) \ge 0 \tag{B}$$

From (A) and (B) it follows that $f(\alpha) = 0$.

Since $\alpha \in [a, b]$ and $f(a) < 0, f(b) > 0$ it follows that $\alpha \in (a, b)$.

Thus $\alpha = c$ and the theorem is proved. □

Theorem 6.4 (Intermediate Value Theorem). *Let $g : [a, b] \to \mathbb{R}$ be a continuous function, and let λ be a real number between $g(a)$ and $g(b)$. Then there exists a point $c \in (a, b)$ such that*

$$g(c) = \lambda.$$

Proof. Without loss of generality, assume that $g(a) < \lambda < g(b)$. Define

$$f(x) = g(x) - \lambda.$$

Since g is continuous on $[a, b]$, so is f. Moreover,

$$f(a) = g(a) - \lambda < 0 \quad \text{and} \quad f(b) = g(b) - \lambda > 0.$$

Hence, by Theorem 6.3, there exists $c \in (a, b)$ such that $f(c) = 0$. Therefore, $g(c) = \lambda$. □

Remark 6.5. *Let $I = [a, b]$ be a closed and bounded interval and $f : [a, b] \to \mathbb{R}$ be such that $f(a) \neq f(b)$ and f attains every value between $f(a)$ and $f(b)$ at least once in (a, b). Still f may not be continuous on $[a, b]$.*

For example, let $f : [0, 2] \to \mathbb{R}$ be defined by $f(0) = 0, f(2) = 2$ and

$$f(x) = \begin{cases} x, & 0 < x \le 1, \\ 3 - x, & 1 < x < 2. \end{cases}$$

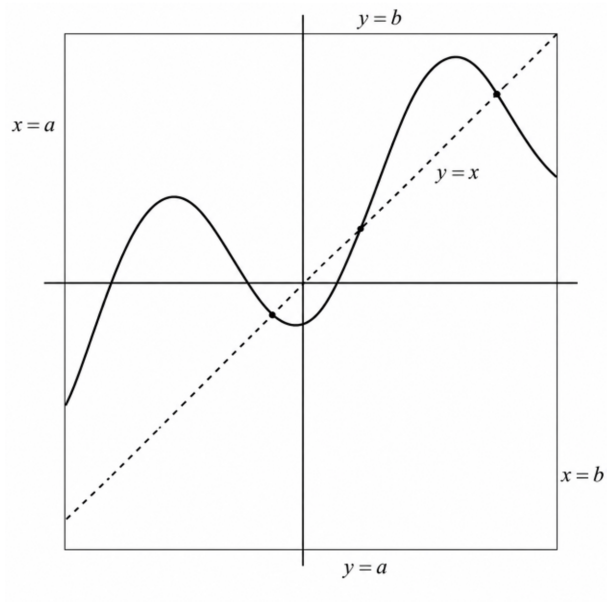
f assumes every value between 0 and 2 on $[0, 2]$. But f is not continuous on $[0, 2]$ since f is not continuous at 1 and 2.

Theorem 6.6 (Fixed Point Theorem). *Let $f : [a, b] \to [a, b]$ be a continuous function. Then there exists $c \in [a, b]$ such that $f(c) = c$. Such a point c is called a **fixed point** of f.*

Proof. Define $g(x) = f(x) - x$. Since g is continuous on $[a, b]$, we have $g(a) = f(a) - a \ge 0$ and $g(b) = f(b) - b \le 0$. If either $g(a) = 0$ or $g(b) = 0$, then we are done. Otherwise, $g(a) > 0 > g(b)$. Hence, by the Intermediate Value Theorem, there exists $c \in (a, b)$ such that $g(c) = 0$. Therefore, $f(c) = c$. □

Example 6.7.

1. *Show that the equation $xe^x = 1$ has a solution in the interval $(0, 1)$.*
2. *Let $f : [0, 2\pi] \to \mathbb{R}$ be a continuous function satisfying $f(0) = f(2\pi)$. Show that there exists $x \in [0, \pi]$ such that $f(x) = f(x + \pi)$.*



3. *Show that every real polynomial of odd degree defines an onto function from $\mathbb{R}$ to $\mathbb{R}$.*
Solution.

1. *Define $f : [0,1] \to \mathbb{R}$ by $f(x) = xe^x - 1$. Since f is continuous on $[0,1]$, we have $f(0) = -1 < 0$ and $f(1) = e - 1 > 0$. Hence, by the Intermediate Value Theorem, there exists $c \in (0,1)$ such that $f(c) = 0$. Therefore, $ce^c = 1$.*

2. *Define $g : [0,\pi] \to \mathbb{R}$ by $g(x) = f(x) - f(x+\pi)$. Since f is continuous, so is g. Moreover,*

$$g(0) = f(0) - f(\pi) \quad \text{and} \quad g(\pi) = f(\pi) - f(2\pi) = f(\pi) - f(0) = -g(0).$$

If $g(0) = 0$, then $f(0) = f(\pi)$ and we are done. Otherwise, $g(0)$ and $g(\pi)$ have opposite signs. Hence, by the Intermediate Value Theorem, there exists $c \in (0,\pi)$ such that $g(c) = 0$. Therefore, $f(c) = f(c+\pi)$.

3. *Let $p : \mathbb{R} \to \mathbb{R}$ be a polynomial of odd degree, and let $y \in \mathbb{R}$ be arbitrary. We wish to show that there exists $x \in \mathbb{R}$ such that $p(x) = y$. Define the function $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = p(x) - y$. Since f is continuous, it suffices to show that f changes sign. As p has odd degree, we have $\lim_{x\to\infty} p(x) = \infty$ and $\lim_{x\to-\infty} p(x) = -\infty$, or vice versa, depending on the sign of the leading coefficient. Hence, there exist $a, b \in \mathbb{R}$ such that $f(a) < 0 < f(b)$. By the Intermediate Value Theorem, there exists $c \in (a,b)$ such that $f(c) = 0$. Therefore, $p(c) = y$. Since y was arbitrary, $p : \mathbb{R} \to \mathbb{R}$ is onto.*

Definition 6.1. *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. We say that f is **bounded on** A if there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.*

Example 6.8. 1. *The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x$ is not bounded. Indeed, for every $M > 0$, choosing $x = M + 1$ gives $f(x) = M + 1 > M$. Hence, f is unbounded.*

2. *The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \sin x$ is bounded. In fact, $|f(x)| \leq 1$ for every $x \in \mathbb{R}$.*

Theorem 6.9 (Weierstrass Theorem). *Let $f : [a,b] \to \mathbb{R}$ be a continuous function. Then f is bounded.*

Sketch of the proof. The proof is based on the same least upper bound argument used in the proof of the Intermediate Value Theorem.

Consider the set of all points $x \in [a, b]$ such that f is bounded on the interval $[a, x]$. Since f is continuous, it is bounded in a neighborhood of a. Thus, this set is nonempty and bounded above by b. Let c be its least upper bound.

If $c < b$, then continuity of f at c implies that f is bounded in a neighborhood of c. Consequently, f is bounded on a larger interval $[a, c_1]$ for some $c_1 > c$, contradicting the definition of c. Hence, $c = b$, and therefore f is bounded on $[a, b]$. □

Theorem 6.10 (Extreme Value Theorem). *Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Then there exist points $x_{\min}, x_{\max} \in [a, b]$ such that*

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for every $x \in [a, b]$.

Proof. By the Weierstrass Theorem, the set $A = \{f(x) : x \in [a, b]\}$ is bounded. Let $M = \sup A$.

For each $n \in \mathbb{N}$, choose $x_n \in [a, b]$ such that $M - \frac{1}{n} < f(x_n) \leq M$. Then $f(x_n) \to M$. Since (x_n) is a bounded sequence, the Bolzano-Weierstrass Theorem yields a subsequence (x_{n_k}) converging to some $x \in [a, b]$. By continuity, $f(x_{n_k}) \to f(x)$. Hence, $f(x) = M$.

Similarly, if $m = \inf A$, there exists $y \in [a, b]$ such that $f(y) = m$. Therefore, f attains both its maximum and minimum values on $[a, b]$. □

Example 6.11.

1. *Show that there is no continuous function $f : [0, 1] \to (0, \infty)$ which is onto.*
2. *Show that there is no continuous function $f : [a, b] \to (0, 1)$ which is onto.*

Solution.

1. *Suppose such a function exists. Since f is continuous on the closed interval $[0, 1]$, the Weierstrass Theorem implies that f is bounded. Hence there exists $M > 0$ such that $f(x) \leq M$ for every $x \in [0, 1]$. Therefore, no number greater than M belongs to the range of f, which contradicts the assumption that f is onto.*
2. *Suppose such a function exists. By the Extreme Value Theorem, there exist $x_{\min}, x_{\max} \in [a, b]$ such that*
$$f(x_{\min}) \leq f(x) \leq f(x_{\max}), \qquad \forall x \in [a, b].$$
Since the range of f is $(0, 1)$, we have $0 < f(x_{\min})$ and $f(x_{\max}) < 1$. Hence, neither 0 nor 1 belongs to the range of f, contradicting the assumption that f is onto.

6.3 Integral Test for Series

Let $f : [a, b] \to \mathbb{R}$ be a continuous function satisfying $\alpha \le f(x) \le \beta$ for all $x \in [a, b]$. Then

$$\alpha(b-a) \le \int_a^b f(x)\,dx \le \beta(b-a).$$

The above estimate has a simple geometric interpretation when f is non-negative: the area under the graph of f lies between the areas of the rectangles of heights α and β over the interval $[a, b]$.

Theorem 6.12 (Integral Test)**.** *Let $f : [1, \infty) \to [0, \infty)$ be a continuous and decreasing function. Suppose that $a_n = f(n)$ for all $n \in \mathbb{N}$. Then:*

1. *If the improper integral $\int_1^\infty f(x)\,dx$ converges, then the series $\sum\limits_{n=1}^{\infty} a_n$ converges.*
2. *If the improper integral $\int_1^\infty f(x)\,dx$ diverges, then the series $\sum\limits_{n=1}^{\infty} a_n$ diverges.*

Proof. For each $n \ge 2$, since f is decreasing, we have

$$a_n \le \int_{n-1}^{n} f(x)\,dx \le a_{n-1}.$$

Summing these inequalities from $k = 2$ to n, we obtain

$$\sum_{k=2}^{n} a_k \le \int_1^n f(x)\,dx \le \sum_{k=1}^{n-1} a_k.$$

Suppose first that $\int_1^\infty f(x)\,dx$ converges. Then the sequence $b_n = \int_1^n f(x)\,dx$ is bounded. Since

$$\sum_{k=2}^{n} a_k \le b_n,$$

the sequence of partial sums $(\sum\limits_{k=2}^{n} a_k)$ is bounded above. As it is also increasing, it converges. Hence $\sum\limits_{n=1}^{\infty} a_n$ converges.

Conversely, suppose that $\int_1^\infty f(x)\,dx$ diverges. Then $b_n \to \infty$. Since

$$b_n \le \sum_{k=1}^{n-1} a_k,$$

the sequence of partial sums of $\sum\limits_{n=1}^{\infty} a_n$ is unbounded. Therefore, the series diverges. $\square$

Example 6.13 (p-Series)**.** *Consider the series $\sum\limits_{n=1}^{\infty} \frac{1}{n^p}$, where $p > 0$. Applying the Integral Test to the function $f(x) = x^{-p}$, we obtain*

$$\int_1^\infty \frac{1}{x^p}\,dx = \begin{cases} \frac{1}{p-1}, & \text{if } p > 1, \\ \infty, & \text{if } 0 < p \le 1. \end{cases}$$

Hence,

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if $p > 1$ and diverges if $0 < p \leq 1$.

In particular, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, whereas $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Example 6.14. *Show that the series $\sum_{n=2}^{\infty} \frac{\log n}{n^p}$ converges for $p > 1$.*

Solution. *Consider the function $f(x) = \frac{\log x}{x^p}$, defined for $x \geq 2$. For $p > 1$, the function f is positive, continuous, and decreasing for all sufficiently large x. By the Integral Test, it is enough to examine*

$$\int_2^{\infty} \frac{\log x}{x^p}\, dx.$$

Using integration by parts with $u = \log x$ and $dv = x^{-p}\, dx$, we obtain

$$\int \frac{\log x}{x^p}\, dx = \frac{x^{1-p} \log x}{1-p} - \frac{x^{1-p}}{(1-p)^2} + C.$$

Since $p > 1$, we have $x^{1-p} \to 0$ and $x^{1-p} \log x \to 0$ as $x \to \infty$. Hence $\int_2^{\infty} \frac{\log x}{x^p}\, dx$ converges. Therefore, by the Integral Test, $\sum_{n=2}^{\infty} \frac{\log n}{n^p}$ converges.

Exercise 6.1. *1. Determine whether the following limits exist. If a limit exists, find its value and justify your answer:*

(a) $\lim_{x \to 2} (2x + x^2)^x$. *(b)* $\lim_{x \to \infty} x(1 + \sin x)$.

2. Find the limits of the following:

(a) $\lim_{x \to 0} \frac{x^2 \sin(1/x)}{\sin x}$ *(b)* $\lim_{x \to 0} (1 + 2x)^{\frac{(x+3)}{x}}$

3. Show that if $f : \mathbb{R} \to \mathbb{R}$ is a continuous function whose range is contained in $\mathbb{Z}$, then f is constant.

4. Let $f : [0, 2] \to \mathbb{R}$ be continuous such that $f(0) = f(2)$. Then there exists $x_1, x_2 \in [0, 2]$ such that

$$x_1 - x_2 = 1 \qquad \text{and} \qquad f(x_1) = f(x_2).$$

5. Find the value of c for which the function

$$f(x) = \begin{cases} x^2 + c, & \text{if } x \leq 1, \\ 2x + 3, & \text{if } x > 1 \end{cases}$$

is continuous at $x = 1$.

6. Let $[a, b]$ be a closed and bounded interval and a function $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. If

$$\sup_{x \in [a,b]} f(x) \neq \inf_{x \in [a,b]} f(x)$$

and μ be a real number lying between

$$\sup_{x\in[a,b]} f(x) \quad \textit{and} \quad \inf_{x\in[a,b]} f(x),$$

then there is a point p in (a,b) such that $f(p)=\mu$.

7 Differentiation

Continuity tells us that a function has no sudden "jumps," but it does not describe the rate at which the function changes. This is the role of derivatives. They answer the question:

How fast does $f(x)$ change as x changes?

Think of driving a car. Distance traveled is a function of time, while the speedometer shows how quickly the distance is changing at that exact moment. This is exactly what a derivative represents: the **rate of change at a given instant**.

In other words, a derivative tells us how much the output changes when the input changes by a tiny amount. Mathematically, this idea is expressed using a **limit**, which gives the precise definition of the derivative.

7.1 Differentiability of Functions

Definition 7.1. *Let (a,b) be an interval and $f:(a,b)\to\mathbb{R}$ be a function. Let $c\in(a,b)$. We say f is* ***differentiable at*** *c if*

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}$$

exists. This limit is called the ***derivative*** *of f at c and is denoted as $f'(c)$.*

Corresponding to a function f, we define its derivative f' as the function whose domain consists of all points x at which the derivative of f exists.

Example 7.1. *1. Let $f:\mathbb{R}\to\mathbb{R}$ be defined by $f(x)=3x+e$. Then f is differentiable at every point of $\mathbb{R}$. For any $c\in\mathbb{R}$, we compute*

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}=\lim_{x\to c}\frac{3x+e-(3c+e)}{x-c}=\lim_{x\to c}\frac{3(x-c)}{x-c}=3.$$

Thus, the derivative exists and $f'(c)=3$, for all $c\in\mathbb{R}$.

2. Let

$$f(x)=\begin{cases} x, & \text{if } x<0,\\ x+1, & \text{if } x\geq 0.\end{cases}$$

Here f is not differentiable at $x=0$. First note $f(0)=0+1=1$. Consider the difference quotient

$$\frac{f(x)-f(0)}{x-0}=\frac{f(x)-1}{x}, \qquad x\neq 0.$$

Now for $x>0$, $f(x)=x+1$, hence

$$\lim_{x\to 0^+}\frac{f(x)-1}{x}=\lim_{x\to 0^+}\frac{x+1-1}{x}=1.$$

If $x < 0$, $f(x) = x$, hence
$$\frac{f(x) - 1}{x} = \frac{x - 1}{x} = 1 - \frac{1}{x}.$$
As $x \to 0^-$, we have $1/x \to -\infty$, so
$$\lim_{x \to 0^-} \frac{f(x) - 1}{x} = +\infty,$$
hence, the left-hand limit diverges. Therefore, the limit does not exist, so f is not differentiable at $x = 0$.

Theorem 7.2. *Let (a, b) be an open interval and a function $f : (a, b) \to \mathbb{R}$ be differentiable at a point $c \in (a, b)$. Then f is continuous at c.*

Proof. For all $x \in (a, b)$, but $x \neq c$,
$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c).$$
Since f is differentiable at c,
$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$
exists and the limit is finite. Also, since $\lim_{x \to c}(x - c) = 0$, we have
$$\lim_{x \to c}[f(x) - f(c)] = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \cdot (x - c) = f'(c) \cdot 0 = 0,$$
since $f'(c)$ is finite. Therefore
$$\lim_{x \to c} f(x) = f(c)$$
and this shows that f is continuous at c. $\square$

Remark 7.3. *The continuity of f at a point $c \in (a, b)$ does not ensure differentiability of f at c. For example, let $f(x) = |x|, x \in \mathbb{R}$. At $x = 0$, $f(x) = 0$. Also f is continuous at 0.*
$$\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} \frac{x}{x} = 1$$
$$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} \frac{-x}{x} = -1.$$
Since the left-hand and right-hand derivatives of f are different, hence f is not differentiable at 0. Hence continuity at a point c does not imply differentiability at c.

Example 7.4. *1. Let*
$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$
Then f is differentiable at 0 and $f'(0) = 0$.

Indeed,
$$\lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{x^2 \sin(1/x)}{x} = \lim_{x \to 0} x \sin(1/x) = 0.$$
But
$$f'(x) = 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right), \qquad x \neq 0.$$
Since $\cos(1/x)$ oscillates as $x \to 0$, the derivative has no limit at 0. Hence f' is not continuous at 0.

2. *Consider*
$$f(x) = \begin{cases} x + x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$$
Then f is differentiable on $\mathbb{R}$ with
$$f'(0) = 1,$$
but
$$f'(x) = 1 + 2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right), \qquad x \neq 0.$$
Again, f' oscillates near 0 and hence is not continuous there.

Theorem 7.5 (Algebra of Differentiation). *Let $f, g : (a,b) \to \mathbb{R}$ be differentiable at $c \in (a,b)$, and let $\lambda \in \mathbb{R}$. Then:*

1. ***(Linearity)***
$$(f+g)'(c) = f'(c) + g'(c), \qquad (\lambda f)'(c) = \lambda f'(c).$$

2. ***(Product rule)***
$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

3. ***(Quotient rule)*** *If $g(c) \neq 0$, then*
$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Theorem 7.6 (Chain Rule). *Let $f : (a,b) \to \mathbb{R}$ be differentiable at $c \in (a,b)$, and let g be defined on an interval containing $f((a,b))$ and differentiable at $f(c)$. Then the composition $g \circ f$ is differentiable at c, and*
$$(g \circ f)'(c) = g'(f(c))f'(c).$$

Example 7.7. 1. *Let $f(x) = x^2$ and $g(y) = \sin y$. Then*
$$(g \circ f)(x) = \sin(x^2).$$
By the Chain Rule,
$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = \cos(x^2) \cdot (2x) = 2x\cos(x^2).$$

2. *Let $f(x) = e^x$ and $g(y) = \ln y$, defined on $(0, \infty)$. Then*
$$(g \circ f)(x) = \ln(e^x) = x.$$
By the Chain Rule,
$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = \frac{1}{e^x} \cdot e^x = 1,$$
which matches the fact that the derivative of x is 1.

3. *Let $f(x) = \sqrt{x}$ on $(0, \infty)$ and $g(y) = y^3$. Then*
$$(g \circ f)(x) = (\sqrt{x})^3 = x^{3/2}.$$
By the Chain Rule,
$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x) = 3(\sqrt{x})^2 \cdot \frac{1}{2\sqrt{x}} = 3x \cdot \frac{1}{2\sqrt{x}} = \frac{3}{2}\sqrt{x}.$$

7.2 Applications

Suppose a function starts and ends at the same height. Do you expect its slope to be zero at some point in between? The answer is affirmative. This remarkable result is known as Rolle's Theorem. The theorem is named after the French mathematician Michel Rolle, who proved the result for polynomials in 1691 without using differential calculus. A rigorous proof in the modern setting was later given by Augustin-Louis Cauchy in 1823.

Theorem 7.8 (**Rolle's Theorem**). *Let $f : [a, b] \to \mathbb{R}$ be a function satisfying:*

1. *f is continuous on $[a, b]$*
2. *f is differentiable on (a, b)*
3. *$f(a) = f(b)$*

Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof. Since f is continuous on $[a, b]$, f is bounded on $[a, b]$. Let

$$\sup_{x\in[a,b]} f(x) = M, \qquad \inf_{x\in[a,b]} f(x) = m.$$

By the property of continuity there exists a point c in $[a, b]$ such that $f(c) = M$ and there exists point d in $[a, b]$ such that $f(d) = m$. Two cases arise.

Case 1. $M = m$.

In this case $f(x) = M$ for all $x \in [a, b]$. Therefore $f'(x) = 0$ for all $x \in [a, b]$. The theorem holds trivially in this case.

Case 2. $M \neq m$.

In this case at least one of M and m, if not both, must be unequal to $f(a)$ (and $f(b)$). Let $M \neq f(a)$. Then $c \neq a, c \neq b$. $\therefore\ a < c < b$. By condition (ii), $f'(c)$ exists. If possible, let $f'(c) > 0$. Then

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} > 0.$$

Hence there exists a positive δ such that

$$\frac{f(x) - f(c)}{x - c} > 0$$

for all $x \in [a, b]$ satisfying $0 < |x - c| < \delta$. This implies $f(x) - f(c) > 0$ for all $x \in [a, b]$ satisfying $c < x < c + \delta$. i.e., $f(x) > M$ for all $x \in [a, b]$ satisfying $c < x < c + \delta$. This contradicts that M is the supremum of f on $[a, b]$. Consequently, $f'(c) \neq 0 \ldots$ (i) If possible, let $f'(c) < 0$. Then

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} < 0.$$

Hence there exists a positive δ such that

$$\frac{f(x) - f(c)}{x - c} < 0$$

for all $x \in [a, b]$ satisfying $0 < |x - c| < \delta$. This implies $f(x) > f(c)$ for all $x \in [a, b]$ satisfying $c - \delta < x < c$. i.e., $f(x) > M$ for all $x \in [a, b]$ satisfying $c - \delta < x < c$. This contradicts that M is the supremum of f on $[a, b]$. Consequently, $f'(c) \neq 0 \ldots$ (ii)

From (i) and (ii) it follows that $f'(c) = 0$. Therefore $\xi = c$. If however, $M = f(a) = f(b)$, then $m \neq f(a)$ and therefore $d \neq a, d \neq b$ and $a < d < b$. Proceeding with similar arguments we can prove $f'(d) = 0$. Therefore $\xi = d$. This completes the proof of the theorem. □

Remark 7.9. 1. *The conditions of Rolle's theorem are sufficient, but not necessary. Consider*

$$f(x) = |x| + |x-1|, \qquad x \in [-1, 2].$$

Then

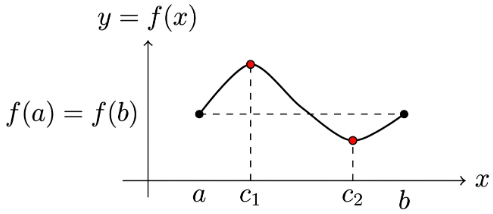
$$f(x) = \begin{cases} 1-2x, & -1 \le x < 0, \\ 1, & 0 \le x \le 1, \\ 2x-1, & 1 < x \le 2. \end{cases}$$

The function f is continuous on $[-1, 2]$, but is not differentiable at $x = 0, 1$. Also, $f(-1) = f(2) = 3$. Thus, f does not satisfy all the conditions of Rolle's theorem. However,

$$f'(x) = 0, \qquad 0 < x < 1.$$

Hence, $f'(\xi) = 0$ for some $\xi \in (-1, 2)$, even though the hypotheses of Rolle's theorem are not all satisfied.

2. *Geometrically, the theorem guarantees the existence of a point where the tangent to the curve $y = f(x)$ is horizontal. Physically, if a particle returns to its initial position, then at some intermediate time its instantaneous velocity must be zero.*



Example 7.10. 1. *The hypotheses of Rolle's Theorem are sufficient but not necessary. That is, a function may satisfy the conclusion of the theorem even if one or more of its hypotheses fail. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3$. On the interval $[-1, 1]$, we have $f(-1) \neq f(1)$, so the hypothesis $f(a) = f(b)$ of Rolle's Theorem is not satisfied. Nevertheless,*

$$f'(x) = 3x^2,$$

and hence $f'(0) = 0$. Thus, the conclusion of Rolle's Theorem holds even though one of its hypotheses fails.

2. *Consider the equation $x^3 + x + 1 = 0$. Suppose $a \neq b$ are two solutions. By Rolle's Theorem, there exists $c \in (a, b)$ such that*

$$3c^2 + 1 = 0,$$

which is impossible. Hence the equation has at most one real solution.

Theorem 7.11 (Mean value theorem (Lagrange)). *Let a function $f : [a, b] \to \mathbb{R}$ be such that (i) f is continuous on $[a, b]$, and (ii) f is differentiable at every point of (a, b).*

Then there exists at least a point ξ in (a, b) such that

$$\frac{f(b) - f(a)}{b - a} = f'(\xi).$$

Proof. Let us define $\phi : [a, b] \to \mathbb{R}$ by

$$\phi(x) = f(x) + Ax, \quad x \in [a, b] \text{ where } A \text{ is a constant.}$$

Clearly, ϕ is continuous on $[a, b]$, since f is continuous on $[a, b]$; and ϕ is differentiable at every point of (a, b), since f is differentiable at every point of (a, b). Let us choose A such that $\phi(a) = \phi(b)$. Then $f(a) + Aa = f(b) + Ab$ and this determines

$$A = \frac{f(b) - f(a)}{a - b}.$$

For this choice of A, ϕ satisfies all conditions of Rolle's theorem on $[a, b]$. Therefore there exists at least a point ξ in (a, b) such that $\phi'(\xi) = 0$. But $\phi'(\xi) = f'(\xi) + A$ and therefore

$$0 = f'(\xi) + \frac{f(b) - f(a)}{a - b},$$

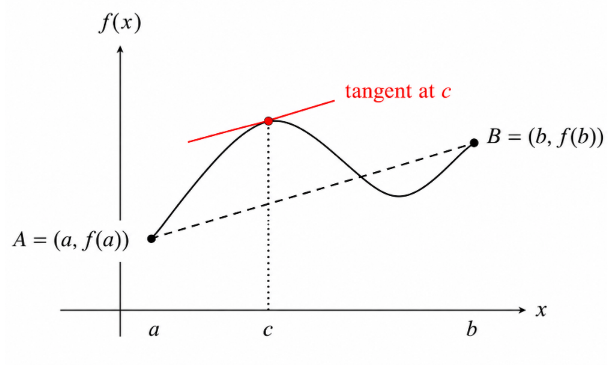
or,

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

□

Remark 7.12. *Geometrically, it means that the tangent to the curve at $x = c$ is parallel to the secant line joining $(a, f(a))$ and $(b, f(b))$, that is,*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Example 7.13. 1. *Show that*

$$|\sin x - \sin y| \leq |x - y|$$

for all $x, y \in \mathbb{R}$.

2. *Show that for every $x > 0$,*

$$\ln(1 + x) < x.$$

3. *Show that*

$$|e^x - e^y| \leq e^{\max\{x,y\}}|x - y|$$

for all $x, y \in \mathbb{R}$ has at most one real solution.

4. *Suppose $f : J \to \mathbb{R}$ is differentiable on an interval J and*

$$f'(x) = 0 \quad \textit{for all } x \in J.$$

Show that f is constant on J.

5. *Let* $f : [2, 5] \to \mathbb{R}$ *be continuous and differentiable on* $(2, 5)$*. Assume that*

$$f'(x) = (f(x))^2 + \pi \quad \text{for all } x \in (2, 5).$$

True or false: $f(5) - f(2) = 3$.

6. *Let* $f : (0, 1] \to \mathbb{R}$ *be differentiable and suppose that*

$$|f'(x)| < 1 \quad \text{for all } x \in (0, 1].$$

Define

$$a_n = f\left(\frac{1}{n}\right).$$

Show that the sequence (a_n) *converges.*

7. *Suppose that* $f : J \to \mathbb{R}$ *is differentiable on an interval* J *and satisfies*

$$0 < m \le f'(x) \le M, \qquad \forall x \in J,$$

where $m, M > 0$*. Show that for every* $x, y \in J$ *with* $x < y$*,*

$$m(y - x) \le f(y) - f(x) \le M(y - x).$$

Solution.

1. *Assume* $x \neq y$*. By the Mean Value Theorem, there exists* c *between* x *and* y *such that*

$$\sin x - \sin y = \cos(c)(x - y).$$

Taking absolute values,

$$|\sin x - \sin y| = |\cos(c)||x - y| \le |x - y|.$$

2. *Apply the Mean Value Theorem to* $f(t) = \ln t$ *on* $[1, 1+x]$*. Then there exists* $c \in (1, 1+x)$ *such that*

$$\ln(1 + x) - \ln 1 = \frac{1}{c}x.$$

Since $c > 1$*, we have* $\frac{1}{c} < 1$*. Hence*

$$\ln(1 + x) = \frac{x}{c} < x.$$

3. *By the Mean Value Theorem, there exists* c *between* x *and* y *such that*

$$e^x - e^y = e^c(x - y).$$

Since $e^c \le e^{\max\{x,y\}}$*, the result follows.*

4. *Let* $x, y \in J$ *with* $x < y$*. Since* J *is an interval,* $[x, y] \subseteq J$*. By the Mean Value Theorem, there exists* $c \in (x, y)$ *such that*

$$f(y) - f(x) = f'(c)(y - x).$$

Since $f'(c) = 0$*, it follows that*

$$f(y) = f(x).$$

As x *and* y *are arbitrary,* f *is constant on* J*.*

5. *The statement is false. By the Mean Value Theorem, there exists* $c \in (2,5)$ *such that*
$$f(5) - f(2) = f'(c)(5-2) = 3f'(c).$$
Since
$$f'(c) = (f(c))^2 + \pi > \pi,$$
we obtain
$$f(5) - f(2) > 3\pi > 3.$$
Hence the given statement is false.

6. *By the Mean Value Theorem, for each* $n \geq 1$*, there exists* $c_n \in \left(\frac{1}{n+1}, \frac{1}{n}\right)$ *such that*
$$a_{n+1} - a_n = f\left(\frac{1}{n+1}\right) - f\left(\frac{1}{n}\right) = f'(c_n)\left(\frac{1}{n+1} - \frac{1}{n}\right).$$
Hence,
$$|a_{n+1} - a_n| \leq \left|\frac{1}{n+1} - \frac{1}{n}\right| = \frac{1}{n(n+1)}.$$
Therefore, for $m > n$*,*
$$|a_m - a_n| \leq \sum_{k=n}^{m-1} |a_{k+1} - a_k| \leq \sum_{k=n}^{m-1} \frac{1}{k(k+1)} = \frac{1}{n} - \frac{1}{m} \leq \frac{1}{n}.$$
Since $\frac{1}{n} \to 0$*, the sequence* (a_n) *is Cauchy in* $\mathbb{R}$ *and so it converges.*

7. *By the Mean Value Theorem, there exists* $c \in (x,y)$ *such that*
$$f(y) - f(x) = f'(c)(y-x).$$
Since
$$m \leq f'(c) \leq M$$
and
$$y - x > 0,$$
we obtain
$$m(y-x) \leq f(y) - f(x) \leq M(y-x).$$

Theorem 7.14 (Constant Function Theorem). *Let* $f : [a,b] \to \mathbb{R}$ *be continuous on* $[a,b]$ *and differentiable on* (a,b)*. If*
$$f'(x) = 0 \qquad \text{for all } x \in (a,b),$$
then f *is constant on* $[a,b]$*.*

Proof. Let $x_1, x_2 \in [a,b]$ and $a \leq x_1 < x_2 \leq b$. Then f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2). By the Mean Value Theorem, there exists a point $\xi \in (x_1, x_2)$ such that
$$f'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$
But $f'(\xi) = 0$ by hypothesis. Therefore,
$$f(x_2) = f(x_1).$$
Since x_1 and x_2 are arbitrary points in $[a,b]$, f is constant on $[a,b]$. $\square$

Corollary 7.15. *Let $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ be both continuous on $[a, b]$ and differentiable on (a, b). If*

$$f'(x) = g'(x) \qquad \text{for all } x \in (a, b),$$

then $f = g + c$, where $c \in \mathbb{R}$ is a constant.

Theorem 7.16 (Monotonicity Theorem). *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then:*

1. *If $f'(x) \geq 0$, for all $x \in (a, b)$, then f is monotone increasing on $[a, b]$.*
2. *If $f'(x) \leq 0$, for all $x \in (a, b)$, then f is monotone decreasing on $[a, b]$.*

Proof. 1. Let $x_1, x_2 \in [a, b]$ and $a \leq x_1 < x_2 \leq b$. Then f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2). By the Mean Value Theorem, there exists a point $\xi \in (x_1, x_2)$ such that

$$f'(\xi) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Since $f'(\xi) \geq 0$, and $x_2 - x_1 > 0$, we obtain

$$f(x_2) - f(x_1) \geq 0.$$

i.e.,

$$f(x_2) \geq f(x_1).$$

As x_1 and x_2 are arbitrary in $[a, b]$, f is a monotone increasing function on $[a, b]$.

2. The proof is similar and is left to the reader.

□

Remark 7.17. *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then*

$$\begin{cases} f'(x) > 0 \quad \text{for all } x \in (a, b) \implies f \text{ is strictly increasing on } [a, b], \\ f'(x) < 0 \quad \text{for all } x \in (a, b) \implies f \text{ is strictly decreasing on } [a, b]. \end{cases}$$

Example 7.18.

1. *Prove that $\frac{2x}{\pi} < \sin x$, for $0 < x < \frac{\pi}{2}$.*
2. *Prove that*
$$\left(1 + \frac{1}{x}\right)^x > \left(1 + \frac{1}{y}\right)^y, \text{ if } x, y \in \mathbb{R} \text{ and } x > y > 0.$$
3. *Let $f : \mathbb{R} \to \mathbb{R}$ satisfy*
$$|f(x) - f(y)| \leq (x - y)^2, \text{ for all } x, y \in \mathbb{R}.$$
Prove that f is a constant function on $\mathbb{R}$.

7.3 Solution.

1. *Let $f(x) = \frac{\sin x}{x}$, $0 < x < \frac{\pi}{2}$. Then*
$$f'(x) = \frac{x \cos x - \sin x}{x^2}.$$
Since $x < \tan x$, $0 < x < \frac{\pi}{2}$, we have $x \cos x - \sin x < 0$. Hence $f'(x) < 0$, and therefore f is strictly decreasing on $(0, \pi/2)$. Thus
$$f(x) > f\left(\frac{\pi}{2}\right) = \frac{2}{\pi}.$$
Therefore, $\frac{2x}{\pi} < \sin x$, $0 < x < \frac{\pi}{2}$.

2. *Let*

$$f(x) = \left(1 + \frac{1}{x}\right)^x, \qquad x > 0.$$

Then

$$f'(x) = \left(1 + \frac{1}{x}\right)^x \left[\log\left(1 + \frac{1}{x}\right) - \frac{1}{1+x}\right].$$

Define

$$\phi(x) = \log(1+x) - \frac{x}{1+x}, \qquad x \geq 0.$$

Then

$$\phi'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} > 0 \qquad (x > 0).$$

Hence ϕ *is strictly increasing on* $[0, \infty)$. *Since*

$$\phi(0) = 0,$$

we obtain

$$\log(1+x) > \frac{x}{1+x}, \qquad x > 0.$$

Replacing x *by* $1/x$ *gives*

$$\log\left(1 + \frac{1}{x}\right) > \frac{1}{1+x}, \qquad x > 0.$$

Therefore,

$$f'(x) > 0, \qquad x > 0.$$

Thus f *is strictly increasing on* $(0, \infty)$. *Hence, if* $x > y > 0$,

$$\boxed{\left(1 + \frac{1}{x}\right)^x > \left(1 + \frac{1}{y}\right)^y}.$$

3. *By the given condition,*

$$-(x-y)^2 \leq f(x) - f(y) \leq (x-y)^2.$$

Fix $c \in \mathbb{R}$ *and put* $x = c + h$ *and* $y = c$. *Then*

$$-h^2 \leq f(c+h) - f(c) \leq h^2.$$

If $h \neq 0$, *division by* h *gives*

$$-h \leq \frac{f(c+h) - f(c)}{h} \leq h \qquad (h > 0),$$

and

$$h \leq \frac{f(c+h) - f(c)}{h} \leq -h \qquad (h < 0).$$

In either case, by the Squeeze Theorem,

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = 0.$$

Hence

$$f'(c) = 0.$$

Since $c \in \mathbb{R}$ *was arbitrary,*

$$f'(x) = 0 \qquad \textit{for all } x \in \mathbb{R}.$$

By the Constant Function Theorem, f *is constant on* $\mathbb{R}$.

Theorem 7.19 (Cauchy's Mean Value Theorem). *Let f and g be continuous real-valued functions on $[a,b]$ which are differentiable in (a,b). Then there is a point $x \in (a,b)$ at which*

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x).$$

Proof. Left as an exercise. $\square$

Theorem 7.20 (Darboux). *Let $I = [a,b]$ and a function $f : I \to \mathbb{R}$ be differentiable on I. Let $f'(a) \neq f'(b)$. If k be a real number lying between $f'(a)$ and $f'(b)$ then there exists a point c in (a,b) such that $f'(c) = k$.*

Example 7.21. *1. Let $f : [-1,1] \to \mathbb{R}$ be defined by*

$$f(x) = \begin{cases} 0, & x \in [-1,0] \\ 1, & x \in (0,1]. \end{cases}$$

Does there exist a function g such that

$$g'(x) = f(x), \quad x \in [-1,1]?$$

If possible, let there exist a function $g : [-1,1] \to \mathbb{R}$ such that

$$g'(x) = f(x), \quad x \in [-1,1].$$

Then g is differentiable on $[-1,1]$ and

$$g'(x) = \begin{cases} 0, & x \in [-1,0] \\ 1, & x \in (0,1]. \end{cases}$$

Since g is differentiable on $[-1,1]$ and $g'(-1) \neq g'(1)$, by Darboux's theorem g' must assume every real number lying between $g'(-1)$ and $g'(1)$, i.e., between 0 and 1 on $[-1,1]$. But this is not so and therefore g does not exist.

Theorem 7.22. *If $f : [a,b] \to \mathbb{R}$ be differentiable on $[a,b]$ then the derived function f' cannot have a jump discontinuity on $[a,b]$.*

7.4 Local Extrema

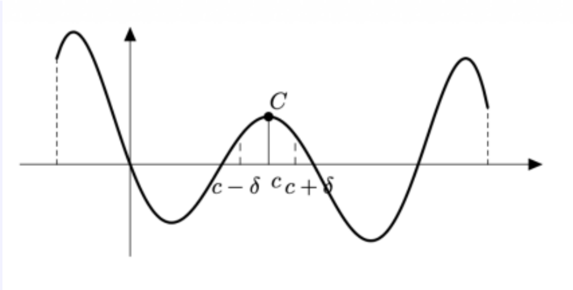
In the real world, systems fluctuate between peaks and valleys rather than moving in straight lines. Studying local maxima and minima allows us to locate these critical turning points where a system reaches an optimal high or low state. In business, they help maximize profit and minimize cost under realistic constraints. In machine learning, algorithms navigate error landscapes to find local minima and improve prediction accuracy. In physics, objects naturally settle at local energy minima for structural stability. Mathematically, these points mark where the rate of change is zero ($f'(c) = 0$), turning abstract curves into practical, optimized decisions.

Definition 7.2 (Local Maximum and Local Minimum)**.** *Let $J \subseteq \mathbb{R}$ be an interval and let $f : J \to \mathbb{R}$ be a function. A point $c \in J$ is called a* ***local maximum*** *of f if there exists $\delta > 0$ such that*

$$(c - \delta, c + \delta) \subseteq J$$

and

$$f(x) \leq f(c) \qquad \text{for all } x \in (c - \delta, c + \delta).$$



Similarly, $c \in J$ is called a ***local minimum*** *of f if there exists $\delta > 0$ such that*

$$(c - \delta, c + \delta) \subseteq J$$

and

$$f(x) \geq f(c) \qquad \textit{for all } x \in (c - \delta, c + \delta).$$

A point c is called a ***local extremum*** *if it is either a local maximum or a local minimum.*

Definition 7.3 (Global Maximum and Global Minimum)**.** *Let $J \subseteq \mathbb{R}$ be an interval and let $f : J \to \mathbb{R}$ be a function.*

A point $x_0 \in J$ is called a ***global maximum*** *of f on J if*

$$f(x) \leq f(x_0) \quad \textit{for all } x \in J.$$

Similarly, $x_0 \in J$ is called a ***global minimum*** *of f on J if*

$$f(x) \geq f(x_0) \quad \textit{for all } x \in J.$$

Do you believe that there is any relation between a global maximum and a local maximum? For example, is any global maximum necessarily a local maximum? Or is any local maximum necessarily a global maximum?

Example 7.23. *Consider the following functions on $\mathbb{R}$:*

1. ***Local maximum.*** *Let $f(x) = -x^2$. At $c = 0$, we have $f(0) = 0$. For x close to 0, $f(x) = -x^2 \leq 0 = f(0)$. Hence, $c = 0$ is a local maximum point.*
2. ***Local minimum.*** *Let $f(x) = x^2$. At $c = 0$, we have $f(0) = 0$. For x close to 0, $f(x) = x^2 \geq 0 = f(0)$. Hence, $c = 0$ is a local minimum point.*
3. ***Saddle point.*** *Let $f(x) = x^3$. At $c = 0$, we have $f(0) = 0$. But in every neighborhood of 0, there are points $x > 0$ where $f(x) > 0 = f(0)$ and points $x < 0$ where $f(x) < 0 = f(0)$. Also $f'(c) = 0$. So $c = 0$ is neither a local maximum nor a local minimum; it is a saddle point.*

Example 7.24. *Consider the function $f : [a, b] \to \mathbb{R}$ defined by $f(x) = x$. Since $f(x) \leq b = f(b)$ for every $x \in [a, b]$, the point b is a global maximum of f. However, b is not a local maximum, as it is an endpoint of the interval. Similarly, a is a global minimum of f, but it is not a local minimum.*

Theorem 7.25 (Fermat's Theorem)**.** *Let $J \subseteq \mathbb{R}$ be an interval and let $f : J \to \mathbb{R}$ be differentiable at $c \in J$. If c is a local extremum of f, then*

$$f'(c) = 0.$$

Proof. Assume that c is a local maximum of f. Then, for h sufficiently small,

$$f(c+h) \le f(c).$$

Hence,

$$\frac{f(c+h)-f(c)}{h} \le 0 \quad \text{if } h>0,$$

and

$$\frac{f(c+h)-f(c)}{h} \ge 0 \quad \text{if } h<0.$$

Passing to the limit as $h \to 0^+$ and $h \to 0^-$, we obtain

$$f'(c) \le 0 \quad \text{and} \quad f'(c) \ge 0.$$

Therefore,

$$f'(c) = 0.$$

The proof for a local minimum is analogous. □

Example 7.26. *The assumption that c is an interior point is essential in Fermat's Theorem. If c lies on the boundary of the domain, the conclusion $f'(c)=0$ may fail. For instance, let*

$$f:[0,\infty)\to\mathbb{R}, \qquad f(x)=x.$$

At $c=0$, we have $f(0)=0$, and clearly

$$f(x) \ge f(0) \quad \textit{for all } x\in[0,\infty),$$

so f has a local minimum at 0. However, $c=0$ is not an interior point of the domain. Moreover, $f'(0)=1\neq 0$, which shows that the conclusion of Fermat's Theorem does not hold in this case.

Theorem 7.27 (First derivative test for extrema). *Let f be continuous on $I=[a,b]$ and c be an interior point of I. Let f be differentiable on (a,c) and (c,b).*

1. *If there exists a neighbourhood $(c-\delta, c+\delta)\subset I$ such that*

$$f'(x)\ge 0 \quad \textit{for } x\in(c-\delta,c), \qquad f'(x)\le 0 \quad \textit{for } x\in(c,c+\delta),$$

then f has a local maximum at c.

2. *If there exists a neighbourhood $(c-\delta, c+\delta)\subset I$ such that*

$$f'(x)\le 0 \quad \textit{for } x\in(c-\delta,c), \qquad f'(x)\ge 0 \quad \textit{for } x\in(c,c+\delta),$$

then f has a local minimum at c.

3. *If $f'(x)$ keeps the same sign on $(c-\delta,c)$ and $(c,c+\delta)$, then f has no extremum at c.*

Proof. 1. Let $x\in(c-\delta,c)$. Applying the Mean Value Theorem to f on $[x,c]$, there exists $\xi\in(x,c)$ such that

$$f(c)-f(x)=(c-x)f'(\xi).$$

Since $f'(\xi)\ge 0$, we have

$$f(x)\le f(c), \qquad x\in(c-\delta,c).$$

Let $x\in(c,c+\delta)$. Applying the Mean Value Theorem to f on $[c,x]$, there exists $\eta\in(c,x)$ such that

$$f(x)-f(c)=(x-c)f'(\eta).$$

Since $f'(\eta) \le 0$, we have
$$f(x) \le f(c), \qquad x \in (c, c+\delta).$$
It follows that
$$f(x) \le f(c) \qquad \text{for all } x \in N(c,\delta) \cap I.$$
Therefore f has a local maximum at c.

2. Similar proof.

3. Let $f'(x) > 0$ for $x \in (c-\delta, c)$ and $x \in (c, c+\delta)$. Then
$$f(x) < f(c) \quad \text{for } x \in (c-\delta, c),$$
and
$$f(c) < f(x) \quad \text{for } x \in (c, c+\delta).$$
Therefore f has a local minimum at c. Similar proof if $f'(x) < 0$ on both sides of c.

□

Remark 7.28. *The converse of the theorem is not true. For example, let*
$$f(x) = 2x^2 + x^2 \sin\frac{1}{x}, \qquad x \ne 0, \qquad f(0) = 0.$$
Then f has a local minimum at 0.

For $x \ne 0$,
$$f'(x) = 4x + 2x\sin\frac{1}{x} - \cos\frac{1}{x}.$$
Thus $f'(x)$ takes both positive and negative values on both sides of 0 in every neighbourhood of 0.

Example 7.29. *1. Let $f(x) = |x|$, $x \in \mathbb{R}$. f is continuous on $\mathbb{R}$, but is not differentiable at 0. Moreover,*
$$f'(x) < 0 \quad \text{for } x \in (-\delta, 0), \qquad f'(x) > 0 \quad \text{for } x \in (0, \delta)$$
for some $\delta > 0$. Therefore f has a local minimum at 0.

2. Let $f(x) = |x-1| + |x-2|$, $x \in [0,3]$. Then
$$f(x) = \begin{cases} 3-2x, & 0 \le x < 1, \\ 1, & 1 \le x \le 2, \\ 2x-3, & 2 < x \le 3. \end{cases}$$
f is continuous on $[0,3]$, but is not differentiable at 1 and 2. Also
$$f'(x) < 0 \quad \text{for } x \in (0,1), \qquad f'(x) = 0 \quad \text{for } x \in (1,2),$$
and
$$f'(x) > 0 \quad \text{for } x \in (2,3).$$
Therefore f has a local minimum at 1 and at 2.

3. *Let* $f(x) = (x-1)^2(x-3)^3,\ x \in \mathbb{R}$. *Then*

$$\begin{aligned} f'(x) &= 2(x-1)(x-3)^3 + 3(x-1)^2(x-3)^2 \\ &= (x-1)(x-3)^2(5x-9). \end{aligned}$$

Hence $f'(x) = 0$ *at* $x = 1,\ 3,\ \frac{9}{5}$. *For* $x \in (1-\delta, 1)$, $f'(x) > 0$, *while for* $x \in (1, 1+\delta)$, $f'(x) < 0$. *Therefore* f *has a local maximum at* 1.

Also, $f'(x) > 0$ *on both sides of* 3 *in a sufficiently small neighbourhood of* 3. *Therefore* f *has neither a maximum nor a minimum at* 3.

Finally,

$$f'(x) > 0 \quad \text{for } x \in \left(\frac{9}{5} - \delta, \frac{9}{5}\right),$$

and

$$f'(x) < 0 \quad \text{for } x \in \left(\frac{9}{5}, \frac{9}{5} + \delta\right).$$

Therefore f *has a local maximum at* $\frac{9}{5}$.

Theorem 7.30 (**Higher order derivative test for extrema**). *Let* $f : I \to \mathbb{R}$ *and* c *be an interior point of* I. *If*

$$f'(c) = f''(c) = \cdots = f^{(n-1)}(c) = 0$$

and

$$f^{(n)}(c) \neq 0,$$

then f *has*

(i) no extremum at c *if* n *is odd, and*

(ii) a local extremum at c *if* n *is even:*

  (a) a local maximum if $f^{(n)}(c) < 0$,

  (b) a local minimum if $f^{(n)}(c) > 0$.

Example 7.31. 1. *Consider* $f(x) = x^5 + 2x^6,\ c = 0$. *We have* $f'(x) = 5x^4 + 12x^5$, *so* $f'(0) = 0$. *Similarly,*

$$f''(0) = f'''(0) = f^{(4)}(0) = 0,$$

while $f^{(5)}(0) = 120 \neq 0$. *Thus the first nonzero derivative is the fifth derivative. Since* $n = 5$ *is odd,* f *has* ***no extremum at*** 0.

2. *Consider* $f(x) = x^4 - 3x^5 + 2x^6,\ c = 0$. *At* $c = 0$, $f'(0) = f''(0) = f'''(0) = 0$, *and* $f^{(4)}(0) = 24 > 0$. *Therefore,* $n = 4$ *is even and* $f^{(4)}(0) > 0$, *so* 0 *is a* ***local minimum***. *Indeed,*

$$f(x) = x^4(1 - 3x + 2x^2).$$

Since

$$1 - 3x + 2x^2 > 0$$

for all sufficiently small x, *we have* $f(x) > 0 = f(0)$ *near* 0. *Hence* 0 *is a local minimum.*

Exercise 7.1. 1. *Show that the continuity hypothesis in Rolle's Theorem cannot be omitted. Consider*

$$f(x) = \begin{cases} x, & \text{if } -1 \le x < 0, \\ 0, & \text{if } x = 0, \\ x, & \text{if } 0 < x \le 1. \end{cases}$$

Verify that $f(-1) = f(1)$ *and that* f *is differentiable on* $(-1, 1)$. *Show that there is no* $c \in (-1, 1)$ *such that* $f'(c) = 0$.

2. *Show that the differentiability hypothesis in Rolle's Theorem cannot be omitted. Consider*

$$f(x) = |x|, \qquad -1 \leq x \leq 1.$$

Verify that $f(-1) = f(1)$ *and that* f *is continuous on* $[-1, 1]$. *Show that there is no* $c \in (-1, 1)$ *such that* $f'(c) = 0$.

3. *Show that the condition* $f(a) = f(b)$ *in Rolle's Theorem cannot be omitted. Consider*

$$f(x) = x, \qquad -1 \leq x \leq 1.$$

Verify that f *is continuous on* $[-1, 1]$ *and differentiable on* $(-1, 1)$. *Show that there is no* $c \in (-1, 1)$ *such that* $f'(c) = 0$.

4. *Let* $g : [-2\pi, 2\pi] \to \mathbb{R}$ *be defined by*

$$g(x) = \cos x.$$

Show that $x = 0$ *is both a local maximum and a global maximum of* g. *What can you say about the points* $x = \pm 2\pi$*?*

5. *Suppose* f *is differentiable on* $[0, 2]$ *with*

$$f'(0) = 7, \qquad f'(2) = 3.$$

For what values of λ *can you guarantee the existence of some* $c \in (0, 2)$ *such that*

$$f'(c) = \lambda?$$